Malware Analysis Report For elf-Linux-SuperH4-bash

Binary Type

ELF (Executable and Linkable Format). (Approved)

Analyzing File

- **File Name**: elf-Linux-SuperH4-bash
- **File Extension**:
- **File Type**: ELF (Executable and Linkable Format).
- **MD5 Hash**: a7f84e65fb2e432c4adb098b076cf4f6
- **SHA256 Hash**: 712d635367f9bfd3a36b86b15046ee7c29d7fb0a69a6d0ce1ac4f42b7d8f3841

File Identification

File detected as a valid Linux executable.

ELF File Analysis

ELF File Analysis: True

ELF File Imports

- libreadline.so.6
- libhistory.so.6
- libncurses.so.5
- libdl.so.2
- libc.so.6

ELF File Exports

No Exported functions were found.

String Analysis

ASCII Strings

- /lib/ld-linux.so.2
- a$01
- WqL_8@
-]aa8$F
- G)`s0&
- }3$MG>
- *l0zxO
- o|>U8*C
- vyd5!t
- R'&6F6
- i OGb&<
- i}=:qW
- libreadline.so.6
- *ITM*deregisterTMCloneTable
- **gmon_start**
- *Jv*RegisterClasses
- *ITM*registerTMCloneTable
- using_history
- rl*make*bare_keymap
- rl*readline*state
- rl*completion*mark*symlink*dirs
- rl*vi*start_inserting
- history_length
- history_get
- rl*prep*term_function
- rl*delete*text
- emacs*standard*keymap
- rl*add*undo
- rl*invoking*keyseqs
- rl*bind*keyseq*in*map
- history_tokenize

- rl*filename*quoting_desired
- sh*unset*nodelay_mode
- rl*tab*insert
- tilde*additional*suffixes
- rl*explicit*arg
- rl*parse*and_bind
- rl_crlf
- remove_history
- rl*vi*bWord
- history_expand
- rl*named*function
- history*search*delimiter_chars
- rl*completer*word*break*characters
- rl*terminal*name
- rl_end
- emacs*meta*keymap
- history*get*time
- rl*function*of_keyseq
- sh*get*env_value
- rl*tilde*expand
- rl*attempted*completion_over
- rl_dispatching
- rl*read*init_file
- vi*movement*keymap
- rl*filename*dequoting_function
- where_history
- rl*completion*type
- rl*directory*completion_hook
- rl*funmap*names
- tilde*expansion*preexpansion_hook
- rl*editing*mode
- rl*last*func
- sh*single*quote
- rl*filename*quoting_function
- history*quotes*inhibit_expansion
- sh*set*lines*and*columns
- history_base
- rl*forward*word
- rl*char*is*quoted*p
- rl*completion*quote_character
- rl_done
- rl_initialize
- rl*bind*key*in*map
- rl*unbind*function*in*map
- rl*completion*entry_function
- rl*add*defun
- rl*completion*suppress_quote
- rl_instream
- rl*num*chars*to*read
- rl*get*previous_history
- rl*set*keymap
- history*subst*char
- rl*completion*append_character
- rl*push*macro_input
- rl*ignore*some*completions*function
- rl*variable*bind
- write_history
- rl*unbind*key*in*map
- history*inhibit*expansion_function
- rl*generic*bind
- unstifle_history
- rl*attempted*completion_function
- rl*forced*update_display
- rl*completion*invoking_key
- rl*bind*key*if*unbound*in*map
- sh*get*home_dir
- rl*sort*completion_matches
- rl*filename*rewrite_hook
- rl*variable*dumper
- rl_outstream
- rl*list*funmap_names

- history*is*stifled
- rl*bind*keyseq
- rl*completion*mode
- rl*executing*keymap
- rl*completer*quote_characters
- rl*filename*completion_desired
- emacs*ctlx*keymap
- free*history*entry
- rl_insert
- rl*read*key
- history*write*timestamps
- rl*get*keymap*by*name
- rl*filename*completion_function
- tilde*additional*prefixes
- history*comment*char
- rl*complete*internal
- rl*special*prefixes
- rl*line*buffer
- rl*on*new_line
- rl*username*completion_function
- rl_newline
- rl_complete
- rl*vi*editing_mode
- rl*function*dumper
- rl*get*keymap
- rl*insert*text
- vi*insertion*keymap
- rl*variable*value
- rl*readline*name
- rl*completion*matches
- rl*startup*hook
- history*truncate*file
- rl_redisplay
- read_history
- rl*basic*word*break*characters
- rl_point
- rl*menu*completion*entry*function
- rl*macro*dumper
- rl*yank*last_arg
- history*set*pos
- replace*history*entry
- rl*reset*terminal
- rl*vi*end_word
- rl_mark
- rl*set*screen_size
- rl*kill*text
- rl*completion*found_quote
- read*history*range
- rl*deprep*term_function
- history*max*entries
- history*expansion*char
- rl*filename*quote_characters
- rl*directory*rewrite_hook
- rl*menu*complete
- rl_ding
- rl*get*termcap
- rl*completion*suppress_append
- libhistory.so.6
- libncurses.so.5
- libdl.so.2
- dlopen
- dlclose
- dlerror
- libc.so.6
- setuid
- strcasestr
- socket
- fflush
- strcpy
- *_printf*chk
- setgrent

- pathconf
- *IO*putc
- setlocale
- mbrtowc
- towlower
- gai_strerror
- strsignal
- strncmp
- dcngettext
- setservent
- strrchr
- _*longjmp*chk
- regexec
- dcgettext
- wcrtomb
- getpwuid
- _*fdelt*chk
- closedir
- strncpy
- sigprocmask
- regfree
- getgrent
- unlink
- putchar
- select
- towupper
- getpid
- strftime
- iswalnum
- iswlower
- strtod
- _*ctype*get*mb*cur_max
- endpwent
- strtol
- isatty
- strchrnul
- getppid
- wcschr
- iswctype
- setitimer
- sigemptyset
- getaddrinfo
- memset
- strstr
- _*errno*location
- tcsetattr
- confstr
- endgrent
- mbsnrtowcs
- mbtowc
- ttyname
- clearerr
- unsetenv
- __fxstat64
- _*fprintf*chk
- sigaddset
- iswupper
- putenv
- wcswidth
- __sigsetjmp
- stdout
- getrusage
- _*strtoll*internal
- strnlen
- memcpy
- fclose
- tcsetpgrp
- _*memmove*chk
- wcscmp
- _*vsnprintf*chk
- strtoul
- killpg
- strcat

- strcasecmp
- setpgid
- getpeername
- mbsinit
- _*strncpy*chk
- getgid
- __lxstat64
- opendir
- __xstat64
- _*ctype*b_loc
- regcomp
- stderr
- sigdelset
- setrlimit64
- group_member
- _*snprintf*chk
- getgroups
- getuid
- readlink
- getegid
- __fpurge
- strncasecmp
- fileno
- gethostname
- getcwd
- fwrite
- gettimeofday
- sigaction
- geteuid
- _*memcpy*chk
- waitpid
- localtime
- lseek64
- strchr
- _*vfprintf*chk
- fdopen
- readdir64
- tcgetattr
- _*ctype*toupper_loc
- _*strcpy*chk
- _*ctype*tolower_loc
- freeaddrinfo
- setvbuf
- wcscoll
- getrlimit64
- _*sprintf*chk
- wcsdup
- getdtablesize
- memmove
- endservent
- fopen64
- bindtextdomain
- wcsrtombs
- setgid
- faccessat
- _*strtoull*internal
- _*asprintf*chk
- tcgetpgrp
- _*libc*start_main
- ferror
- stpcpy
- wcslen
- strcoll
- wcsncmp
- sysconf
- getservent
- wctomb
- __environ
- _*fpscr*values
- _edata
- _*bss*start
- sigwinch_sighandler
- sh_modcase

- parse_shellopts
- parse_string
- protected_mode
- _*lib*ccsu_fini
- search*for*command
- xdupmbstowcs
- shell*glob*filename
- clean_itemlist
- declare_doc
- count*all*jobs
- command*word*completion_function
- array*dequote*escapes
- find_variable
- running_trap
- break_doc
- jobs_doc
- dirspell
- coproc_fdrestore
- fmtulong
- shopt_setopt
- array*dispose*element
- set_shellopts
- pop_args
- lastpipe_opt
- no*exit*on*failed*exec
- should*ignore*glob_matches
- duplicate*buffered*stream
- bashline_reset
- getifs
- executing_builtin
- all_digits
- tempvar_list
- completions*to*stringlist
- get*current*prompt_level
- pat_subst
- add*or*supercede*exported*var
- show*name*attributes
- top*level*mask
- bash*initialize*history
- the*printed*command
- add*unwind*protect
- var_lookup
- fd_ispipe
- no*symbolic*links
- strvec*from*word_list
- expand*string*unsplit*to*string
- reset*mail*timer
- the*printed*command*except*trap
- set*default*locale
- make*while*command
- assoc_patsub
- executing*line*number
- check*dev*tty
- reset*shell*options
- buffered_ungetchar
- yynerrs
- cleanup*the*pipeline
- wmatchlen
- dispose*function*def_contents
- restricted_shell
- terminate_immediately
- hash_doc
- zwrite
- array_shift
- signal*is*trapped
- bash*tilde*expand
- ulimit_builtin
- logout_doc
- get*dirstack*from_string
- zmapfd
- select_doc
- dot*or*dotdot

- wdcache
- gather*here*documents
- print_rlimtype
- dstack
- trim_pathname
- run*debug*trap
- close*fd*bitmap
- break_builtin
- sv_histtimefmt
- close*buffered*stream
- bash*getcwd*errstr
- breaking
- history*line*sin_file
- glob_star
- convert*var*to_array
- bash_copyright
- subshell_envp
- remember*mail*dates
- fg_doc
- builtin_error
- builtin_address
- remove_backslashes
- free*mail*files
- parser_error
- make*variable*value
- command_errstr
- coproc_fdchk
- mailstat
- expand*words*no_vars
- u32cconv
- it_arrayvars
- reader_loop
- local*exported*variables
- sh_syntabsiz
- coproc_flush
- make_command
- local_builtin
- hostname*list*initialized
- find*path*file
- assoc_insert
- file_exists
- zcatfd
- caller_doc
- expand*words*shellexp
- which*set*flags
- ifs_firstc
- strcreplace
- caller_builtin
- bind*keyseq*to*unix*command
- execute*variable*command
- builtin_warning
- exit*immediately*on_error
- find*string*in_alist
- push_context
- sigint_sighandler
- primary_prompt
- mbschr
- redirection_expand
- hash_builtin
- shell_initialized
- tt_setnocanon
- fg*percent*doc
- bash_logout
- close*buffered*fd
- ttrestore
- ifs_cmap
- disown_builtin
- find*or*make*array*variable
- array*remove*quoted_nulls
- sh_stat
- continue_builtin
- glob*ignore*case

- strlist_resize
- coproc_setvars
- bgpids
- disable*priv*mode
- falarm
- array_rshift
- shell*version*string
- xbcopy
- strlist_flush
- posix*readline*initialize
- print*arith*command
- test_command
- suspend_builtin
- fnx_tofs
- is*basic*table
- coproc_fdsave
- expand*prompt*string
- grouping*braces*doc
- xtrace_reset
- set*sigint*handler
- no*empty*command_completion
- base_pathname
- alloc*word*desc
- default*buffered*input
- addtimeval
- make*coproc*command
- array*expand*index
- alias*expand*word
- secondary_prompt
- array_modcase
- lists*stopped*jobs
- string*list*dollar_star
- pop*var*context
- enable*hostname*completion
- push_scope
- maybe*make*export_env
- array*shift*element
- copy*function*def_contents
- push_stream
- get*minus*o_opts
- echo_builtin
- sh_optopt
- buffered_getchar
- return_EOF
- current*fd*s to_close
- bg_doc
- zgetline
- exit_builtin
- let_doc
- skip*to*delim
- signal*is*special
- notify*and*cleanup
- make_word
- builtin_doc
- dot*found*in_search
- termsig_sighandler
- read_command
- ttfd_cbreak
- build_version
- exit_doc
- saved*command*line_count
- current*command*line_count
- copy_command
- expand*word*unsplit
- do*word*assignment
- find*shell*builtin
- make*bare*word
- setup*glob*ignore
- read*secondary*line
- shell*eof*token
- dircomplete*expand*relpath
- current*readline*line

- initialize_readline
- copy*function*def
- popd_doc
- yyerror
- get*local*var
- unset*bash*input
- return*catch*value
- assoc*dequote*escapes
- yyparse
- wait*for*job
- sv*strict*posix
- array_patsub
- make*if*command
- it_running
- times_doc
- legal*alias*name
- sh_regmatch
- current*command*first*line*saved
- dispose*function*def
- all*visible*functions
- subshell_level
- parse_bashopts
- bash*badsub*errmsg
- complete_builtin
- list_string
- ungetc*with*restart
- set*signal*handler
- already*making*children
- last*made*pid
- z*read*intr
- word_split
- find*special*builtin
- sh_optind
- wait*for*background_pids
- wordexp_only
- sh_eaccess
- shell*compatibility*level
- quote*string*for_globbing
- sv_mail
- with*input*from_string
- lcurrent
- make*local*array_variable
- bind*int*variable
- set*login*shell
- pushed*string*list
- ifs_var
- find*user*command
- sh_realpath
- fmtumax
- array*to*assign
- pcomp_curcmd
- nohup_job
- set*working*directory
- last*command*exit_signal
- shell*is*restricted
- make*function*def
- times_builtin
- shell_tty
- sh*backslash*quote
- strlist_print
- save*bash*input
- set_signal
- maybe*set*sigchld_trap
- initialize*shell*options
- hash_dispose
- get*name*for_error
- find*index*in_alist
- readonly_builtin
- assign*array*element
- describe_pid
- unwind*protect*mem
- get*working*directory

- set*if*not
- fail*glob*expansion
- localetrans
- assign*array*from_string
- sv_optind
- return_doc
- sv_opterr
- terminate*stopped*jobs
- dircomplete_expand
- set*default*locale_vars
- sv_funcnest
- sv*comp*wordbreaks
- command*string*index
- sh*backslash*quote*for*double_quotes
- pcomp_curcs
- ttnocanon
- sh_nojobs
- until_doc
- set_pwd
- find*function*def
- phash_remove
- gen*compspec*completions
- source_builtin
- kill*all*local_variables
- named*function*string
- tt_setonechar
- set*pipestatus*array
- get*numeric*arg
- internal_warning
- hash_copy
- expand*string*to_string
- tt_setcbreak
- list*minus*o_opts
- find*tempenv*variable
- subshell*top*level
- sh_erange
- pop_context
- with*input*from_stream
- expand_string
- assoc*to*assign
- return_catch
- it_jobs
- array_quote
- netopen
- string*list*dollar_at
- it_keywords
- array_dispose
- glob_vector
- change_flag
- trap_doc
- unset_builtin
- initialize*shell*builtins
- dequote_list
- bind_function
- unbind_func
- expand*word*leave_quoted
- set*signal*ignored
- bashline_reinitialize
- print_command
- parse*symbolic*mode
- xtrace_fp
- initialize_signals
- string*rest*of_args
- signal_names
- xrealloc
- ps2_prompt
- EOF_Reached
- sv_terminal
- make*new*array_variable
- word*token*list
- check_identifier
- make*until*command

- command_connect
- bash*add*history
- ttfd_noecho
- the*printed*command_size
- unquoted*glob*pattern_p
- reset*shopt*options
- push*var*context
- make*cond*command
- sccs_version
- sh_syntaxtab
- prog_completes
- programming_error
- save*pipestatus*array
- substring
- job*exit*signal
- new*var*context
- hash_string
- unlink*fifo*list
- history*lines*this_session
- sh_physpath
- job*exit*status
- progcomp_remove
- sh_mktmpfd
- source*searches*cwd
- set*job*control
- get*word*from_string
- print*var*value
- map_over
- set*sigchld*handler
- reinit*special*variables
- fc*execute*file
- unbind*function*def
- logout_builtin
- print*shift*error
- fifos_pending
- copy*fifo*list
- export_builtin
- open*buffered*stream
- hup*on*exit
- coproc_list
- reset*signal*handlers
- it_users
- push_args
- initialize*shell*variables
- set*impossible*sigchld_trap
- skipsubscript
- tempenv*assign*error
- echo_doc
- phash_flush
- array*from*word_list
- array_copy
- wait_doc
- extract*arithmetic*subst
- xmalloc
- yy*input*name
- kill*current*pipeline
- eval_builtin
- dircomplete_spelling
- convert*var*to_assoc
- hist_verify
- internal_wstrmatch
- maybe*execute*file
- strvec_strcmp
- restart*job*control
- filter_stringlist
- compspec_copy
- compgen_doc
- makunbound
- phash_search
- this*shell*function
- history*expansion*inhibited
- interrupt_immediately

- array_remove
- minus*o*option_value
- sync*buffered*stream
- pwd_builtin
- absolute_program
- progcomp_dispose
- ttfd_onechar
- run*trap*cleanup
- get*group*array
- decode*prompt*string
- mail_warning
- arith*for*doc
- xtrace_init
- static*shell*builtins
- set*current*prompt_level
- make*bare*simple_command
- sh*double*quote
- subshell_argv
- zreset
- current*readline*line_index
- eof_encountered
- umatchlen
- assoc*quote*escapes
- dot_doc
- read*but*dont_execute
- phash_create
- source_file
- set*return*trap
- hash_size
- loop_level
- cdable_vars
- privileged_mode
- ifs_value
- builtin_usage
- executable_file
- expand*assignment*string*to*string
- bind*var*to_int
- shell_name
- remove_alias
- set*directory*hook
- all*array*variables
- give*terminal*to
- just*one*command
- trap*saved*exit_value
- array*create*element
- command_error
- allow*null*glob_expansion
- true_doc
- getcoprocbypid
- assign*array*var*from*word_list
- make*new*assoc_variable
- debugging*login*shell
- get*new*window_size
- sh_mktmpfp
- pipefail_opt
- with*input*from_stdin
- builtin_builtin
- want*pending*command
- ps1_prompt
- interrupt_state
- return_builtin
- display*signal*list
- xtrace_fd
- copy*word*list
- string*list*internal
- compspec_create
- ttsave
- redirection_error
- restore*pgrp*pipe
- set*var*attribute
- complete_doc
- read_doc

- dispose_variable
- variable_context
- run*sigchld*trap
- strlist_create
- coproc_wclose
- _*data*start
- strlist*to*word_list
- make*group*command
- set*minus*o_option
- execute*shell*function
- wait*signal*received
- set*bash*input_fd
- hash_walk
- clear*hostname*list
- wait_builtin
- set*buffered*stream
- strvec*to*word_list
- run*unwind*frame
- dispose_command
- internal_strmatch
- extract*colon*unit
- promptvars
- compspec_dispose
- strlist_walk
- kill_doc
- sh_notbuiltin
- place*keywords*in_env
- expand*string*assignment
- assoc_quote
- do_redirections
- sh_invalidoptname
- extract*command*subst
- disallow*filename*globbing
- initialize*jobs*signals
- print_timeval
- colon_doc
- timeval*to*cpu
- exec_argv0
- it_hostnames
- coproc_init
- set*pipestatus*from_exit
- match*pattern*wchar
- kill_pid
- test_builtin
- hist*last*line_added
- free*pushed*string_input
- set*error*trap
- adjust*shell*level
- update*export*env_inplace
- unbind_variable
- sv_ifs
- pop*dollar*vars
- xtrace_set
- coproc_closeall
- function*trace*mode
- sh_opterr
- it_commands
- xparse_dolparen
- restore*original*signals
- sh_coproc
- brace_expansion
- expand_words
- sub*append*string
- ttgetattr
- ansicstr
- quote*globbing*chars
- init*yy*io
- do_assignment
- reset*mail*files
- zreadc
- shopt_doc
- dump*po*strings

- shift_doc
- parse*and*execute_cleanup
- getmaxgroups
- reset*terminating*signals
- word*list*remove*quoted*nulls
- posix_initialize
- bind_doc
- enable_doc
- compopt_builtin
- set*original*signal
- bash*history*disable
- subshell_environment
- printf_doc
- dispose*fd*bitmap
- without*job*control
- get*array*value
- strip*trailing*ifs_whitespace
- maybe*call*trap_handler
- shell*start*time
- prompt*string*pointer
- typeset_doc
- exec_builtin
- forced_interactive
- malloc*trace*at_exit
- phash_insert
- source*uses*path
- indirection*level*string
- list*all*jobs
- enable_builtin
- *IO*stdin_used
- strvec_len
- stream_list
- sh*un*double_quote
- asynchronous_notification
- delete*all*contexts
- set*locale*var
- patch_level
- signal_name
- make*funcname*visible
- absolute_pathname
- append_process
- get*all*original_signals
- last*asynchronous*pid
- assoc*to*word_list
- yychar
- getopts_doc
- it_aliases
- array*variable*name
- find*variable*internal
- local_doc
- print*array*assignment
- coproc_pidchk
- set*dollar*vars_unchanged
- sh_validfd
- string*to*rlimtype
- command_substitute
- maybe*make*restricted
- function_doc
- err_unboundvar
- it_directories
- sh_badjob
- reset*completer*word*break*chars
- array*quote*escapes
- unbind_args
- coproc_close
- list*rest*of_args
- bash*directory*completion_matches
- expand*arith*string
- current*readline*prompt
- literal_history
- strsub
- interactive_shell

- while_doc
- eof*encountered*limit
- clean*simple*command
- arith_doc
- sh_closepipe
- array*assign*list
- sh_invalidid
- check*hashed*filenames
- sh_exit
- strlist_append
- sys_error
- dispose_redirects
- programmable_completions
- make*command*string
- unfreeze*jobs*list
- fatal_error
- ifs_firstchar
- suspend_doc
- dispose*word*desc
- save*parser*state
- make_absolute
- current*prompt*string
- array_walk
- make*arith*for_command
- getmaxchild
- bash*tilde*find_word
- sh_badpid
- strvec_copy
- sh_getopt
- array*to*word_list
- assoc_flush
- case_doc
- fsleep
- trap*to*sighandler
- mapfile_builtin
- get*variable*value
- dump*translatable*strings
- uinttostr
- ansiexpand
- signal*in*progress
- dispose_words
- get*shopt*options
- compgen_builtin
- assoc_remove
- timeval*to*secs
- dollar*dollar*pid
- get*dollar*var_value
- no*line*editing
- current_token
- binary_test
- assoc*keys*to*word*list
- set*default*lang
- _*libc*csu_init
- need*here*doc
- internal_error
- jobs_builtin
- file_isdir
- run*pending*traps
- ifs*first*clen
- compopt_doc
- full_pathname
- setup*ignore*patterns
- list*one*job
- type_builtin
- match*ignore*case
- pop_scope
- maybe*append*history
- bash*re*edit
- assoc_dispose
- show*all*var_attributes
- make*default*mailpath
- delete*all*jobs

- current*command*subst_pid
- have*unwind*protects
- num_fifos
- it_disabled
- sh_invalidopt
- get*string*value
- eval_doc
- bash*history*reinit
- disown_doc
- strvec_create
- exec_doc
- get_exitstat
- save*input*line_state
- init*job*stats
- mark*modified*vars
- history_reediting
- valid*array*reference
- current_user
- false_builtin
- cdspelling
- string*list*pos_params
- all*exported*variables
- put*command*name*into*env
- sh_invalidsig
- setup*history*ignore
- array_insert
- ttnoecho
- bind_builtin
- current*command*number
- dirs_builtin
- false_doc
- enable*history*list
- array*keys*to*word*list
- it_builtins
- command*execution*string
- set*or*show_attributes
- bind*variable*value
- assoc*remove*quoted_nulls
- set*debug*trap
- current_builtin
- cond*expand*word
- run*return*trap
- posixly_correct
- global_variables
- it_functions
- find*global*variable
- report_error
- sh_openpipe
- xstrmatch
- print*func*list
- check*bash*input
- array_dequote
- remove*quoted*escapes
- make*select*command
- sigwinch_received
- ttattr
- set_ppid
- time*to*check_mail
- xpg_echo
- xtrace*print*word_list
- file_iswdir
- error*trace*mode
- it_services
- trap_list
- list*running*jobs
- let_builtin
- type_doc
- export_doc
- sh_neednumarg
- exec*redirection*undo_list
- progcomp_search
- sv*history*control

- global_command
- wlcache
- assoc*to*string
- jobs*m*flag
- pop_stream
- it_signals
- wcsmatch
- cmd_init
- coproc_checkfd
- strvec_sort
- mk_msgstr
- ttonechar
- find_alias
- alias_expand
- zreadretry
- bind*array*variable
- ulimit_doc
- reset*internal*getopt
- remember*on*history
- glob*error*return
- progcomp_flush
- sh_makepath
- progcomp_insert
- dispose*saved*dollar_vars
- initialize*job*control
- autocd
- restore_pipeline
- it_variables
- colon_builtin
- err_readonly
- xtrace*print*assignment
- fc_builtin
- interactive_comments
- xtrace_fdchk
- run*interrupt*trap
- make*simple*command
- file_status
- list_reverse
- sv_xtracefd
- string*quote*removal
- bash_license
- nohup*all*jobs
- perform*hostname*completion
- clear*unwind*protect_list
- jump*to*top_level
- xtrace*print*for*command*head
- terminate*current*pipeline
- set*tty*state
- comsub*ignore*return
- xtrace*print*arith_cmd
- u32tochar
- xtrace*print*select*command*head
- unalias_doc
- array*variable*part
- no*invisible*vars
- setup*async*signals
- show*shell*version
- funcnest_max
- stream*on*stack
- tilde_initialize
- it_enabled
- bg_builtin
- history_number
- pre*process*line
- readonly_doc
- list_optarg
- initialize_traps
- copy_redirects
- sv_histchars
- umask_builtin
- make*pattern*list
- shopt_builtin

- no_options
- tteightbit
- list_optopt
- delete*all*aliases
- if_doc
- test*bracket*doc
- restore*parser*state
- return*catch*flag
- close*pgrp*pipe
- command*oriented*history
- get*current*user_info
- reap*dead*jobs
- save*token*state
- pcomp*set*readline_variables
- pwd_doc
- sh_mktmpname
- begin*unwind*frame
- strlist_remove
- release_status
- sv_histsize
- isnetconn
- split*at*delims
- array_create
- make*builtin*argv
- set*var*read_only
- zreadcn
- array*unshift*element
- initialize_bashopts
- execute*command*internal
- terminating_signal
- time_doc
- getcoprocbyname
- initialize*bash*input
- umask_doc
- make*subshell*command
- cd_doc
- close*new*fifos
- u32toutf8
- strlist*prefix*suffix
- sv_globignore
- find*reserved*word
- find*token*in_alist
- last*history*line
- parse*and*execute
- print*var*list
- readarray_doc
- ansic_quote
- array*to*argv
- read*from*stdin
- gnu*error*format
- sh_chkwrite
- maybe*save*shell_history
- strvec_flush
- ttsetattr
- make*here*document
- coproc_dispose
- history_doc
- xtrace*print*cond_term
- hashing_enabled
- ttfd_nocanon
- save*pgrp*pipe
- dont*save*function_defs
- builtin*address*internal
- coproc_rclose
- fc_doc
- ignore_signal
- shift_builtin
- strvec_resize
- get*random*number
- bash*history*enable
- make*case*command
- evalexp

- array_keys
- polite*directory*format
- dirs_doc
- declare_builtin
- executing
- fd*is*bash_input
- copy_redirect
- free*trap*strings
- echo*input*at_read
- move*to*high_fd
- strlist_dispose
- make*arith*command
- noglob*dot*filenames
- setifs
- stupidly*hack*special_variables
- shell*function*defs
- show*var*attributes
- match*pattern*char
- this*shell*builtin
- load_history
- check*add*history
- continuing
- stdin_redirects
- fg_builtin
- print*asso*cassignment
- subst*assign*varlist
- termsig_handler
- force_fignore
- exit_shell
- sh_badopt
- sv_hostfile
- array*to*string
- indirection_level
- hash_bucket
- last*command*subst_pid
- strlist_sort
- save_pipeline
- last*shell*builtin
- mbstrlen
- it_exports
- continue_doc
- parse*and*execute_level
- strip_trailing
- initialize_itemlist
- extended_quote
- is_directory
- loptend
- dispose*used*env_vars
- mapfile_doc
- assoc_dequote
- copy_word
- test_unop
- strvec_remove
- array*needs*making
- it_files
- parse_command
- terminal_pgrp
- check*binary*file
- run*unwind*protects
- legal_number
- sh_setlinebuf
- make*word*from_token
- it_stopped
- reset_parser
- brace_expand
- char*is*quoted
- localeexpand
- printf_builtin
- force*append*history
- unset*sigwinch*handler
- remember_args
- set_bashopts

- progcomp_create
- print*cond*command
- expand*compound*array_assignment
- tt_seteightbit
- coproc_doc
- sh_wrerror
- init*mail*dates
- run*exit*trap
- getopts_builtin
- dispose_word
- test_binop
- sh*contains*shell_metas
- top*level*cleanup
- dispose*var*context
- current*host*name
- dispose*cond*node
- this*command*name
- coproc_unsetvars
- sh_restricted
- make*word*flags
- end*job*control
- unbind*array*element
- assoc_modcase
- signal*is*hard_ignored
- set*itemlist*dirty
- strlist*from*word_list
- it_groups
- read_builtin
- get*directory*stack
- list_opttype
- decode_signal
- noclobber
- user*command*matches
- assign*array*var*from*string
- yylval
- input_avail
- do*assignment*no_expand
- chkexport
- subshell_argc
- expand*string*unsplit
- bash*brace*completion
- sv_locale
- set*dirstack*element
- zsyncfd
- running*under*emacs
- pushd_doc
- all*variables*matching_prefix
- sh_optarg
- _getenv
- it_setopts
- mbsmbchar
- wait*intr*buf
- tvlist_ind
- err_badarraysub
- execute_command
- bash*delete*last_history
- same_file
- test_doc
- make*local*variable
- assign*compound*array_list
- check*jobs*at_exit
- read_octal
- delete*all*variables
- assigning*in*environment
- unclosed_pair
- unalias_builtin
- progcomp_walk
- set_lang
- ttcbreak
- get*job*spec
- throw*to*top_level
- describe_command

- assoc_reference
- extract*array*assignment_list
- sourcelevel
- source_doc
- make_redirection
- sort_variables
- sh_mkdoublequoted
- get*job*by_pid
- difftimeval
- strvec_dispose
- set*dollar*vars_changed
- hangup*all*jobs
- find_flag
- executing_list
- ignore*tty*job_signals
- uconvert
- glob*char*p
- push*dollar*vars
- unbound*var*sis_error
- check*window*size
- bash*clear*history
- pipeline_pgrp
- other*token*alist
- bash*input*fd_changed
- initialize_aliases
- print*var*function
- executing*command*builtin
- bash*readline*initialized
- sh_needarg
- hist*last*line_pushed
- have_devfd
- unset_doc
- pushd_builtin
- sv_ignoreeof
- make*for*command
- file_error
- get*alias*value
- no_args
- de_backslash
- original_pgrp
- xtrace*print*case*command*head
- initialize_flags
- echo*command*at_execute
- list_length
- alias*expand*all
- remove*unwind*protect
- array_flush
- number*of*args
- expanding_redir
- delete_job
- trap_builtin
- help_builtin
- get*job*by_name
- sh_ttyerror
- optflags
- command_doc
- debugging_mode
- bash*servicename*completion_function
- restore*input*line_state
- the*current*maintainer
- run*error*trap
- get*hostname*list
- extended_glob
- sv_path
- coproc_alloc
- print*simple*command
- restore*token*state
- make*cond*node
- ansic_shouldquote
- make*word*list
- spname
- get*tty*state

- conditional_doc
- legal_identifier
- variable*help*doc
- stop*making*children
- wait_for
- new*fd*bitmap
- the*current*working_directory
- stop_pipeline
- start_pipeline
- assoc_subrange
- history_expansion
- sv_histignore
- extract*process*subst
- fd*to*buffered_stream
- maybe*add*history
- it_shopts
- last*command*exit_value
- array_slice
- all*shell*variables
- sh_canonpath
- getopts_reset
- executable*or*directory
- history_builtin
- shell_pgrp
- parse*string*to*word*list
- get*group*list
- find_function
- strlist_merge
- bash*dequote*text
- unary_test
- history*delimiting*chars
- array_subrange
- all*visible*variables
- coproc_fdclose
- prog*completion*enabled
- pcomp*set*compspec_options
- progcomp_size
- make_child
- startup_state
- bash*groupname*completion_function
- strlist_copy
- sh_notfound
- zreadcintr
- all*shell*functions
- reset*shell*flags
- make*local*assoc_variable
- coproc_reap
- default*tty*job_signals
- shell_execve
- start_job
- signal*i*signored
- xmbsrtowcs
- num*shell*builtins
- cd_builtin
- popd_builtin
- tt_setnoecho
- global*error*list
- with*input*from*buffered*stream
- stdin_redir
- ttfd_eightbit
- wait*for*single_pid
- free*buffered*stream
- sh_invalidnum
- initialize*terminating*signals
- assign*in*env
- dist_version
- bind*assoc*variable
- map*over*funcs
- connect*async*list
- sh_readonly
- merge*temporary*env
- fnx_fromfs

- bash*delete*histent
- it_bindings
- sh*getopt*restore_state
- unlink_fifo
- restore*default*signal
- bash*default*completion
- restore*pipestatus*array
- expand_aliases
- add_alias
- kill_builtin
- line*number*base
- dispose*exec*redirects
- dequote_string
- shopt_listopt
- discard*unwind*frame
- GLIBC_2.1
- GLIBC_2.0
- GLIBC_2.11
- GLIBC_2.8
- GLIBC_2.15
- GLIBC_2.4
- GLIBC_2.3.4
- GLIBC_2.2
- GLIBC_2.3
- "e2cS3
- x2gsf-F-
- wr#"bl2
- H1!A!A
- "!+@&O
- a#d#e,1
- tCb0`s
- QPc3`}
- 0%CjDc
- 0- c8#
- c,s#f3d
- uPeX%[
- T[HR
- v`fh&S
- s0c8#6
- d@w9+|;
- m@sy-<=8
- r b("m
- wpgx'T
- /Cj"O4
- d|5<5!
- 3 g0fP#
- d|5<5N
- d}C<5R `
- r b(")
- 2aB))!K
- eRd@`!
- 7rfqWx
- G,2{g+b,9
- c<33`]
- 6 V v
- &6(v(V'
- w~VcWx'
- G,2{g*
- w~UvTj
- b,2{g-
- r Q&UX%
- r Q&UX%
- b,2{gH
- r Q&UX%
- G,2{gC
- bre,2+bS`
- G,2{gc
- bre,2+bS`
- w~VcWx'
- w~USWx'
- b,2['q
- r.USWx'
- vr`lUfT

- vr`nUhT
- r Q&UX%
- g,2cP+b
- b,2cP+b
- b,2r&+b
- uaW^U['q
- f^U['q
- b,2{g+b
- G,2{g+b
- /H$"O;
- /Sm"O0
- /Ck"Osi
- #grf`3
- /"O@`+
- .63g>7cbcnsm
- `~3W9z7
- *b:c#`3a
- r"/$c3`
- ChBd@a
- @xsa@q
- /Sl"OT
- /Ch"OBd
- /"O@`/
- 1CcP`/
- n2~3#k8#
- `5#g$c3`:
- \7pb#`:
- /"O@`~
- "b&c@3
- /Ch"Q"O
- /ch"O9
- Cc"TAP
- a"cL131
- e8u %~
- /"ODQEY&1
- "f!W#S"R
- aXqb!T
- 3`;b8;
- /Ch"OBQ
- /Ch"OBQ
- "a2c<1
- d"b(4B!]
- /Ch"OBdH$
- ChDTH$
- 44440488084444488444
- !<3820r.c
- !,2(70w~b
- be['r&
- r c3`=
- @cPb 3
- /Sj"O"(
- /Ch"OBT
- /Ch"OBT
- /Ch"OCT
- /"OBdH$c
- /"OBdH$'
- "a2c81
- 7uR2`!Q
- bch@r9(
- d@r9(,8
- FS"b 3
- BdRe@`Pa
- /X%"OCh
- /X%"OCh
- B/ATH$
- B/ATH$
- /Ci"OSd
- /Ch"Ocd
- /Ch"OSd
- /"OAVh&*
- "OBdH$
- B<2#c)C
- EQDP)!

- e f`5;
- dR&2/b
- EQ)! 1D
- /cl"O>
- C:382#c0s0/d
- a"b 1=
- 80000088000
- g"`Xw#S
- g#SXw"`
- g#SXw"`
- /Cj"O'
- 2`"eBd
- kb+b,A
- c<33`-
- QRf|7s`m
- .6>7x'
- ")AS@R
- r c8#K
- #c\31S2c8#
- /h&"Och
- 2"qS01(
- "OCQ`1
- rbqS"!rR1
- &V'S3`
- /Cj"ON
- rbqS"!rR1
- bbaS"'bR1
- /cj"Ou
- fsa !3f4b
- a8q6!&!
- wg7cc^
- ;c=B#`
- /cn"O{
- a q6!&!
- PDu"/cg
- "OSbRe
- +b-C3`
- f !8vk
- a8qcd6!&!
- +b-C3`
- g,7pa#c
- /H$"OChj
- N2^3X%
- n2~33`
- a q6!&!
- q6!&!\
- r c8#2
- dg3np#
- /Cn"OZ
- wg7cct
- g$w6'&'
- s0c8#+
- /C`"O*
- T@s)-<=
- R!R#`(
- h"c3`*
-)1R0P7[3]4^5Z6.
- a(q6!&!
- fb3g("
- (#czb)#
- F2]n<@
- SdSd"/Q
- /cn"ORbZ
- &cSg8#
- r b("$
- b,2#`=
- f
- r b#`}
- vf#ah&
- f|c<33`m
- r b("T
- T@c3`*
- s0c3`]

- {c=B#`
- +b-C3`
- r b#`~
- r b#`~
- 7vQzR,1
- #r#seS8
- ;c=Gs`
- lwLY(6
- /Sj"OG
- /Ch"OSi(
- R!Q9!01
- g"ePc07
- c"'R!ra
- /Sn"OS
- b@r S"R
- bPr"R#96
- #S"R'3
- 8#Pr SfP2
- b9""!&O
- s`) 3d
- b9""!&O
- /Ch"OS
- "OBjAT
- i"(Bb"!
- /"O@`.
- +b-C3`
- /Sj"Oci
- /y#"O@e
- q"b#1Q
- 7fc0`!
- "a2c31
- "a2c31
- qb!"b("
- #i#e2u\8
- e2u\878
- Sn2~3"(1
- DTCYC`
- `BQc1v
- /cb"O)
- .8>9X%&
- /"OP`(
- .8>9X%C
- "OSich
- /cl"O<
- Sbcar/
- /sn"OBQ
- a0q6!&!
- +b-C3`
- ,5Pc
- gsb,2#`
- gGT(64
- 5Pb,`=@
- U6!&!T
- r b(")
- BdRe@`Pa
- br,2+"*
- +A&O(vK
- f|6``\
- @c0"@a
- sk3i#a
- r b#`n
- B2*s*` se(0!
- w`' gx'
- -b,r+2*")
- +@&O UA
- +@&O UA
- +@&O UA
- +@&O UA
- 3bl2!R 7
- @cPb 3
- q6!&!B
- a(q6!&!
- /Sj"O@`$

- h"ORecmX%
- cPb 3(
- "ORb("
- QQ@1Sa
- /Sd"O[
- CaSd+@
- /cn"Ob
- /"OP`~
- WtV)&06
- /Ch"OSi
- @@ @@ `@` @ `
- /cm"Om
- /"OBb)!
- BQWeY!
- S `#d$eS` =
- b;""!"
- BdRe@`Pa
- g#SXw"`
- /"O@`0
- b,2#`=
- /H$"OChJ
- =@,0W0
- a,q#g7
- V`c8#"
- /Cm"OE
- 2"grWx'
- /Sk"OP
- C:3'3S
- Gz7s2C
- R"d@`-
- R"d@b#`-
- r b("C
- R"d@b#`-
- "d@b#`%
- r bCa("
- /"O@`-
- a2c@q0"
- CaBd+@
- B*2 1m
- 2!TH$&
- 2!TH$&
- S2eX%2
- Q8#+!4
- n2~3"/
- r b(""
- r b("O
- !TSh"O
- 3ffeX%
- !<3820r$
- /Ch"O)
- 2!R"eX%
- gsb,2-
- Au)%R!
- @Z5CVT
- @Z5IVZ
- @Z5AVR
- @Z5KV\
- @Z5EVV
- n2~3"-1
- @Z5MV^
- SQUN2"-^3bd
- /"OP`+
- B!B.2"
- R"b c3`-
- CaAT+@
- /Cj"OH
- Q2dH$s
- b,2#`=
- b,2#`=
- #c<33`}
- b;",c@
- 2!B!B#d"
- \w6'&'P

- /Cn"Oa
- saAQCg
- /ci"O'
- /cn"OZ
- /"O@`/
- RabbQS
- 2aQ"$81
- baRbaS
- /"ObbRaaS,1QR<2W
- <2BeAVU
- (7sbG19
- 287scSg
- !C!CW43ej6
- Gz7x2%
- 7samAx1
- g '3f4bsa
- v b# /
- eTb !("
- ChBdH$
- ChBdH$
- ChBdH$
- X%BQX%
- Bd+@RepVA
- "OBdH$
- /Sn"O)
- w8rpa h
- b
- v`dH$;
- c
- qtc
- /Ch"Oci8
- c
- qtc
- Vsc!W@s`
- e1W2fp43g
- cc@s1U0T@
- S(>#f=
- VHRlS.5
- #qUk#4
- @p1U2V3W4ZL
- L6g[wPfT
- UpV 5
- 2f1Wb(q
- >V?W@sn
- !l6h20r '
- 6l6h20r '
- gfff,kJ
- e]eX%!
- 3fl6b3
- Ch<84`
- CaSd+@
- /cn"O#
- R`QQRR
- @Sa(EF/T0
- /T0T0k
- T0T0T0T0
- aT1C`$@SdT1$@T1$@T1$@T1$@T1$@T1$@
- DV/[eD
- (ET0T0T0
- /T0T0T0
- `T0T0T0T0T0T0T0T0
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- T0T0T0T0T0T0T0T0
- T0T0T0T0T0T0
- ASd$AT0$AT0$AT0$AT0$AT0$AT0$AT0
- 0@P`p
- (08@HPX`hpx
- $(,048<@DHLPTX`dhlptx|
- oDBAAA
- G-C{%-I
- b r("J

- =se9dYf
- x'ccChl
- #g-M-C
- m}C#`[+
-
- Sj3d-Jch
- D3i-HK*-l]
- +'Si3b
- 3g-Jch
- GS`{*.
-
- 2+b#` p
- v5SaCc
- j3z13`&O
- K-AcjS`
- BchS`Cc}H
- d0d0d0d0d0d0d0d0d0d0d0d0d0d0d0d0
- \0X0la
- GNU bash, version %s-(%s)
- sh4-unknown-linux-gnu
- Shell options:
- -irsD or -c command or -O shopt_option
- (invocation only)
- -%s or -o option
- Type `%s -c "help set"' for more information about shell options.
- Type `%s -c help' for more information about shell builtin commands.
- Use the `bashbug' command to report bugs.
- Usage:
- %s [GNU long option] [option] ...
- %s [GNU long option] [option] script-file ...
- GNU long options:
- /usr/share/bashdb/bashdb-main.inc
- run*one*command
- BASH_ENV
- /bin/sh
- I have no name!
- ??host??
- POSIXLY_CORRECT
- POSIX_PEDANTIC
- \s-\v\$
- ~/.bashrc
- %s: option requires an argument
- %s: invalid option
- %c%c: invalid option
- login_shell
- SSH_CLIENT
- SSH2_CLIENT
- /etc/bash/bashrc
- /etc/profile
- ~/.profile
- ~/.bash_profile
- ~/.bash_login
- FUNCNAME
- BASH_SOURCE
- BASH_LINENO
- %s: is a directory
- %s: cannot execute binary file
- debugger
- dump-po-strings
- dump-strings
- init-file
- noediting
- noprofile
- protected
- rcfile
- verbose
- version
- timed out waiting for input: auto-logout
- PROMPT_COMMAND
- reader_loop
- readline stdin
- %a %b %d
- %H:%M:%S

- %l:%M:%S
- %l:%M %p
- unexpected EOF while looking for matching `%c'
- #%^,~:-=?+/
- syntax error near unexpected token `%s'
- syntax error near `%s'
- syntax error: unexpected end of file
- syntax error
- unexpected EOF while looking for matching `)'
- unexpected EOF while looking for `]]'
- syntax error in conditional expression: unexpected token `%s'
- syntax error in conditional expression
- unexpected token `%s', expected`)'
- expected `)'
- unexpected argument `%s' to conditional unary operator
- unexpected argument to conditional unary operator
- unexpected token `%s', conditional binary operator expected
- conditional binary operator expected
- unexpected argument `%s' to conditional binary operator
- unexpected argument to conditional binary operator
- unexpected token `%c' in conditional command
- unexpected token `%s' in conditional command
- unexpected token %d in conditional command
- Use "%s" to leave the shell.
- COPROC
- memory exhausted
- command substitution
- newline
- select
- function
- coproc
- !"#$%&'()*+,-./04
- <@JDA>F?EGHI
- =>>>>??
@@@AAABBCCDDDDDEEEEEEEEEEEFFFFFFFFGGGGHHHHHHIllJJJKK
- OLDPWD
- `%s': not a valid identifier
- /dev/tty
- PROMPT_DIRTRIM
- getcwd: cannot access parent directories
- syntax error: arithmetic expression required
- syntax error: `;' unexpected
- syntax error: `((%s))'
- make*here*document: bad instruction type %d
- here-document at line %d delimited by end-of-file (wanted `%s')
- make_redirection: redirection instruction `%d' out of range
- clean*simple*command
- cprintf: `%c': invalid format character
- <<%s%s
- <<< %s
- {%s}<&%d
- %d<&%d
- {%s}>&%d
- %d>&%d
- {%s}<&%s
- %d<&%s
- {%s}>&%s
- %d>&%s
- {%s}<&%d-
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- %d>&%d-
- {%s}<&%s-
- %d<&%s-
- {%s}>&%s-
- %d>&%s-
- {%s}>&-
- xtrace_set: %d: invalid file descriptor
- xtrace_set: NULL file pointer
- xtrace fd (%d) != fileno xtrace fp (%d)
- %s=(%s)
- for %s in
- select %s in

- case %s in
- case %s in
- for ((
- print_command: bad connector `%d'
- function %s ()
- coproc %s
- print_command
- dispose_command
- cannot duplicate fd %d to fd %d
- eval_builtin
- builtin_env
- return*temp*env
- execute*cond*node
- /dev/null
- cannot redirect standard input from /dev/null: %s
- %s_PID
- %s: %s: bad interpreter
- simple-command
- auto_resume
- substring
- saved redirects
- %s: restricted: cannot specify `/' in command names
- command*not*found_handle
- %s: command not found
- execute_coproc: coproc [%d:%s] still exists
- loop_redirections
- COLUMNS
- %*d%s%s
- execute_connection
- %s: readonly function
- execute_command
- %s: maximum function nesting level exceeded (%d)
- function_calling
- execute-shell-function
- real %2R
- user %2U
- sys %2S
- TIMEFORMAT: `%c': invalid format character
- pipe error
- pipe-file-descriptors
- lastpipe-exec
- execute-command
- %s: %s: cannot open as FILE
- %s: %s: invalid value for trace file descriptor
- OPTERR
- OPTIND
- ignorespace
- ignoredups
- ignoreboth
- erasedups
- HISTFILE
- all*local*variables: no function context at current scope
- make*local*variable: no function context at current scope
- shell-init
- shell level (%d) too high, resetting to 1
- pop*var*context: head of shell_variables not a function context
- pop*var*context: no global_variables context
- pop*scope: head of shell*variables not a temporary environment scope
- BASH_ARGV
- BASH_ARGC
- ~/.sh_history
- ~/.bash_history
- error importing function definition for `%s'
- /usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
- HOSTTYPE
- OSTYPE
- MACHTYPE
- HOSTNAME
- MAILCHECK
- BASH_VERSION
- BASH_VERSINFO
- BASH*EXECUTION*STRING
- IGNOREEOF

- ignoreeof
- HISTCONTROL
- HISTIGNORE
- HISTTIMEFORMAT
- BASH_XTRACEFD
- SECONDS
- BASH_COMMAND
- BASH_SUBSHELL
- RANDOM
- LINENO
- BASHPID
- HISTCMD
- COMP_WORDBREAKS
- DIRSTACK
- GROUPS
- BASH_CMDS
- BASH_ALIASES
- GLOBIGNORE
- PIPESTATUS
- FUNCNEST
- HISTFILESIZE
- HISTSIZE
- HOSTFILE
- LC_ALL
- LC_COLLATE
- LC_CTYPE
- LC_MESSAGES
- LC_NUMERIC
- LC_TIME
- MAILPATH
- TERMCAP
- TERMINFO
- TEXTDOMAIN
- TEXTDOMAINDIR
- histchars
- line
- (null)
- last command: %s
- Aborting...
- warning:
- %s: %s:%s%d:
- %s: %s: %d
- %s: unbound variable
- %s: readonly variable
- bash-maintainers@gnu.org
- unknown command error
- bad command type
- bad connector
- bad jump
- %s%s%s: %s (error token is "%s")
- %s[%s]
- identifier expected after pre-increment or pre-decrement
- missing `)'
- syntax error: operand expected
- exponent less than 0
- division by 0
- `:' expected for conditional expression
- attempted assignment to non-variable
- bug: bad expassign token
- expression recursion level exceeded
- syntax error in expression
- recursion stack underflow
- invalid number
- invalid arithmetic base
- value too great for base
- */%+-&^|
- syntax error: invalid arithmetic operator
-
- Signal %d
- [%d]%c
- Stopped
- Stopped(%s)
- Running

- Done(%d)
- Exit %d
- Unknown status
- (core dumped)
- (wd: %s)
- %s: line %d:
- (core dumped)
- (wd now: %s)
- notify*of*job_status
- start_pipeline: pgrp pipe
- deleting stopped job %d with process group %ld
- [%d] %ld
- describe_pid: %ld: no such pid
- [%ld: %d (%d)] tcsetattr
- SIGCHLD trap
- waitchld: turning on WNOHANG to avoid indefinite block
- initialize*job*control: getpgrp failed
- initialize*job*control: setpgid
- cannot set terminal process group (%d)
- no job control in this shell
- wait_for: No record of process %ld
- wait*for*job: job %d is stopped
- wait: pid %ld is not a child of this shell
- %s: job has terminated
- %s: job %d already in background
- [%d]%s
- (wd: %s)
- fork: retry
- child setpgid (%ld to %ld)
- forked pid %d appears in running job %d
- job-working-directory
- bad substitution: no closing `%s' in %s
- cannot make pipe for command substitution
- cannot make child for command substitution
- command_substitute: cannot duplicate pipe as fd 1
- %s: substring expression < 0
- %s: cannot assign list to array member
- #%^,~:-=?+/}
- #%:-=?+/}
- %s: bad substitution
- $%s: cannot assign in this way
- %s: parameter null or not set
- cannot make pipe for process substitution
- /dev/fd/
- cannot make child for process substitution
- cannot duplicate named pipe %s as fd %d
- process substitution
- bad substitution: no closing "`" in %s
- no match: %s
- unknown
- You have mail in $_
- You have new mail in $_
- The mail in %s has been read
- invalid signal number
- exit trap
- debug trap
- error trap
- return trap
- interrupt trap
- run*pending*traps: bad value in trap_list[%d]: %p
- run*pending*traps: signal handler is SIG_DFL, resending %d (%s) to myself
- trap_handler: bad signal %d
- SIGHUP
- SIGINT
- SIGQUIT
- SIGILL
- SIGTRAP
- SIGABRT
- SIGBUS
- SIGFPE
- SIGKILL
- SIGUSR1
- SIGSEGV

- SIGUSR2
- SIGPIPE
- SIGALRM
- SIGTERM
- SIGSTKFLT
- SIGCHLD
- SIGCONT
- SIGSTOP
- SIGTSTP
- SIGTTIN
- SIGTTOU
- SIGURG
- SIGXCPU
- SIGXFSZ
- SIGVTALRM
- SIGPROF
- SIGWINCH
- SIGPWR
- SIGSYS
- SIGJUNK(32)
- SIGJUNK(33)
- SIGRTMIN
- SIGRTMIN+1
- SIGRTMIN+2
- SIGRTMIN+3
- SIGRTMIN+4
- SIGRTMIN+5
- SIGRTMIN+6
- SIGRTMIN+7
- SIGRTMIN+8
- SIGRTMIN+9
- SIGRTMIN+10
- SIGRTMIN+11
- SIGRTMIN+12
- SIGRTMIN+13
- SIGRTMIN+14
- SIGRTMIN+15
- SIGRTMAX-14
- SIGRTMAX-13
- SIGRTMAX-12
- SIGRTMAX-11
- SIGRTMAX-10
- SIGRTMAX-9
- SIGRTMAX-8
- SIGRTMAX-7
- SIGRTMAX-6
- SIGRTMAX-5
- SIGRTMAX-4
- SIGRTMAX-3
- SIGRTMAX-2
- SIGRTMAX-1
- SIGRTMAX
- RETURN
- cannot reset nodelay mode for fd %d
- cannot allocate new file descriptor for bash input from fd %d
- save*bash*input: buffer already exists for new fd %d
- unwind*frame*run: %s: frame not found
- unwind*frame*discard: %s: frame not found
- argument expected
- %s: integer expression expected
- %s: binary operator expected
- %s: unary operator expected
- `)' expected
- `)' expected, found %s
- missing `]'
- too many arguments
- %s.%d(%d)-%s
- release
- GNU bash, version %s (%s)
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- @(#)Bash version 4.2.45(1) release GNU
- %s: cannot convert indexed to associative array
- array assign
- bad array subscript
- %s[%s: %s
- [%s]: %s
- %s: invalid associative array key
- %s: cannot assign to non-numeric index
- %s: %s: must use subscript when assigning associative array
- ;&()|<>
- %s: cannot create: %s
- %s%s%s
- %s: first non-whitespace character is not `"'
- no closing `%c' in %s
- \"'@<>=;|&()#$`?*[!:{~
- symlink-hook
- completion-ignore-case
- ;|&{(`
- $include
- /etc/hosts
- hostname*completion*file
- C-xC-e
- input-meta
- fc -e "${VISUAL:-${EDITOR:-emacs}}"
- READLINE_LINE
- READLINE_POINT
- bash*execute*unix_command
- bash*execute*unix_command: cannot find keymap for command
- fc -e vi
- fc -e "${VISUAL:-${EDITOR:-vi}}"
- comment-begin
- "'@><=;|&(:
- "'><=;|&(:
- shell-expand-line
- history-expand-line
- magic-space
- shell-forward-word
- shell-backward-word
- shell-kill-word
- shell-backward-kill-word
- history-and-alias-expand-line
- insert-last-argument
- operate-and-get-next
- display-shell-version
- edit-and-execute-command
- complete-into-braces
- complete-filename
- possible-filename-completions
- complete-username
- possible-username-completions
- complete-hostname
- possible-hostname-completions
- complete-variable
- possible-variable-completions
- complete-command
- possible-command-completions
- glob-complete-word
- glob-expand-word
- glob-list-expansions
- dynamic-complete-history
- dabbrev-expand
- ()<>;&|
- *EmptycmD*
- %s: missing colon separator
- FIGNORE
- /usr/share/locale
- setlocale: LC_ALL: cannot change locale (%s)
- setlocale: LC_ALL: cannot change locale (%s): %s
- setlocale: %s: cannot change locale (%s)
- setlocale: %s: cannot change locale (%s): %s
- #: %s:%d
- msgid %s%s
- msgstr ""

- redirection error: cannot duplicate fd
- sh-thd
- file descriptor out of range
- %s: ambiguous redirect
- %s: cannot overwrite existing file
- %s: restricted: cannot redirect output
- cannot create temp file for here-document: %s
- %s: cannot assign fd to variable
- /dev/tcp//
- /dev/udp//
- COMP_LINE
- COMP_POINT
- COMP_TYPE
- COMP_KEY
- COMP_WORDS
- COMP_CWORD
- completion: function `%s' not found
- gen-shell-function-matches
- COMPREPLY
- *DefaultCmD*
- programmable_completion: %s: possible retry loop
- progcomp_insert: %s: NULL COMPSPEC
- %s: cannot allocate %lu bytes (%lu bytes allocated)
- xmalloc
- xrealloc
- Read lines from a file into an array variable.
- A synonym for `mapfile'.
- Read lines from the standard input into an indexed array variable.
- Read lines from the standard input into the indexed array variable ARRAY, or
- from file descriptor FD if the -u option is supplied. The variable MAPFILE
- is the default ARRAY.
- Options:
- -n count
- Copy at most COUNT lines. If COUNT is 0, all lines are copied.
- -O origin
- Begin assigning to ARRAY at index ORIGIN. The default index is 0.
- -s count
- Discard the first COUNT lines read.
- -t
- Remove a trailing newline from each line read.
- -u fd
- Read lines from file descriptor FD instead of the standard input.
- -C callback
- Evaluate CALLBACK each time QUANTUM lines are read.
- -c quantum
- Specify the number of lines read between each call to CALLBACK.
- Arguments:
- ARRAY
- Array variable name to use for file data.
- If -C is supplied without -c, the default quantum is 5000. When
- CALLBACK is evaluated, it is supplied the index of the next array
- element to be assigned and the line to be assigned to that element
- as additional arguments.
- If not supplied with an explicit origin, mapfile will clear ARRAY before
- assigning to it.
- Exit Status:
- Returns success unless an invalid option is given or ARRAY is readonly or
- not an indexed array.
- Modify or display completion options.
- Modify the completion options for each NAME, or, if no NAMEs are supplied,
- the completion currently being executed. If no OPTIONs are given, print
- the completion options for each NAME or the current completion specification.
- Options:
- -o option
- Set completion option OPTION for each NAME
- Change options for the "default" command completion
- Change options for the "empty" command completion
- Using `+o' instead of` -o' turns off the specified option.
- Arguments:
- Each NAME refers to a command for which a completion specification must
- have previously been defined using the `complete' builtin. If no NAMEs
- are supplied, compopt must be called by a function currently generating
- completions, and the options for that currently-executing completion

- generator are modified.
- Exit Status:
- Returns success unless an invalid option is supplied or NAME does not
- have a completion specification defined.
- Display possible completions depending on the options.
- Intended to be used from within a shell function generating possible
- completions. If the optional WORD argument is supplied, matches against
- WORD are generated.
- Exit Status:
- Returns success unless an invalid option is supplied or an error occurs.
- Specify how arguments are to be completed by Readline.
- For each NAME, specify how arguments are to be completed. If no options
- are supplied, existing completion specifications are printed in a way that
- allows them to be reused as input.
- Options:
- -p
- print existing completion specifications in a reusable format
- -r
- remove a completion specification for each NAME, or, if no
- NAMEs are supplied, all completion specifications
- -D
- apply the completions and actions as the default for commands
- without any specific completion defined
- -E
- apply the completions and actions to "empty" commands --
- completion attempted on a blank line
- When completion is attempted, the actions are applied in the order the
- uppercase-letter options are listed above. The -D option takes
- precedence over -E.
- Exit Status:
- Returns success unless an invalid option is supplied or an error occurs.
- Formats and prints ARGUMENTS under control of the FORMAT.
- Options:
- -v var
- assign the output to shell variable VAR rather than
- display it on the standard output
- FORMAT is a character string which contains three types of objects: plain
- characters, which are simply copied to standard output; character escape
- sequences, which are converted and copied to the standard output; and
- format specifications, each of which causes printing of the next successive
- argument.
- In addition to the standard format specifications described in printf(1)
- and printf(3), printf interprets:
- %b
- expand backslash escape sequences in the corresponding argument
- %q
- quote the argument in a way that can be reused as shell input
- %(fmt)T output the date-time string resulting from using FMT as a format
- string for strftime(3)
- Exit Status:
- Returns success unless an invalid option is given or a write or assignment
- error occurs.
- Set and unset shell options.
- Change the setting of each shell option OPTNAME. Without any option
- arguments, list all shell options with an indication of whether or not each
- is set.
- Options:
- -o
- restrict OPTNAMEs to those defined for use with `set -o'
- -p
- print each shell option with an indication of its status
- -q
- suppress output
- -s
- enable (set) each OPTNAME
- -u
- disable (unset) each OPTNAME
- Exit Status:
- Returns success if OPTNAME is enabled; fails if an invalid option is
- given or OPTNAME is disabled.
- Display directory stack.
- Display the list of currently remembered directories. Directories
- find their way onto the list with the `pushd' command; you can get

- back up through the list with the `popd' command.
- Options:
- -c
- clear the directory stack by deleting all of the elements
- -l
- do not print tilde-prefixed versions of directories relative
- to your home directory
- -p
- print the directory stack with one entry per line
- -v
- print the directory stack with one entry per line prefixed
- with its position in the stack
- Arguments:
- +N
- Displays the Nth entry counting from the left of the list shown by
- dirs when invoked without options, starting with zero.
- -N
- Displays the Nth entry counting from the right of the list shown by
- dirs when invoked without options, starting with zero.
- Exit Status:
- Returns success unless an invalid option is supplied or an error occurs.
- Remove directories from stack.
- Removes entries from the directory stack. With no arguments, removes
- the top directory from the stack, and changes to the new top directory.
- Options:
- -n
- Suppresses the normal change of directory when removing
- directories from the stack, so only the stack is manipulated.
- Arguments:
- +N
- Removes the Nth entry counting from the left of the list
- shown by `dirs', starting with zero. For example:` popd +0'
- removes the first directory, `popd +1' the second.
- -N
- Removes the Nth entry counting from the right of the list
- shown by `dirs', starting with zero. For example:` popd -0'
- removes the last directory, `popd -1' the next to last.
- The `dirs' builtin displays the directory stack.
- Exit Status:
- Returns success unless an invalid argument is supplied or the directory
- change fails.
- Add directories to stack.
- Adds a directory to the top of the directory stack, or rotates
- the stack, making the new top of the stack the current working
- directory. With no arguments, exchanges the top two directories.
- Options:
- -n
- Suppresses the normal change of directory when adding
- directories to the stack, so only the stack is manipulated.
- Arguments:
- +N
- Rotates the stack so that the Nth directory (counting
- from the left of the list shown by `dirs', starting with
- zero) is at the top.
- -N
- Rotates the stack so that the Nth directory (counting
- from the right of the list shown by `dirs', starting with
- zero) is at the top.
- dir
- Adds DIR to the directory stack at the top, making it the
- new current working directory.
- The `dirs' builtin displays the directory stack.
- Exit Status:
- Returns success unless an invalid argument is supplied or the directory
- change fails.
- Common shell variable names and usage.
- BASH_VERSION
- Version information for this Bash.
- CDPATH
- A colon-separated list of directories to search
- for directories given as arguments to `cd'.
- GLOBIGNORE
- A colon-separated list of patterns describing filenames to

- be ignored by pathname expansion.
- HISTFILE
- The name of the file where your command history is stored.
- HISTFILESIZE
- The maximum number of lines this file can contain.
- HISTSIZE
- The maximum number of history lines that a running
- shell can access.
- HOME
- The complete pathname to your login directory.
- HOSTNAME
- The name of the current host.
- HOSTTYPE
- The type of CPU this version of Bash is running under.
- IGNOREEOF
- Controls the action of the shell on receipt of an EOF
- character as the sole input. If set, then the value
- of it is the number of EOF characters that can be seen
- in a row on an empty line before the shell will exit
- (default 10). When unset, EOF signifies the end of input.
- MACHTYPE
- A string describing the current system Bash is running on.
- MAILCHECK
- How often, in seconds, Bash checks for new mail.
- MAILPATH
- A colon-separated list of filenames which Bash checks
- for new mail.
- OSTYPE
- The version of Unix this version of Bash is running on.
- PATH
- A colon-separated list of directories to search when
- looking for commands.
- PROMPT_COMMAND
- A command to be executed before the printing of each
- primary prompt.
- PS1
- The primary prompt string.
- PS2
- The secondary prompt string.
- PWD
- The full pathname of the current directory.
- SHELLOPTS
- A colon-separated list of enabled shell options.
- TERM
- The name of the current terminal type.
- TIMEFORMAT
- The output format for timing statistics displayed by the
- `time' reserved word.
- auto_resume
- Non-null means a command word appearing on a line by
- itself is first looked for in the list of currently
- stopped jobs. If found there, that job is foregrounded.
- A value of `exact' means that the command word must
- exactly match a command in the list of stopped jobs. A
- value of `substring' means that the command word must
- match a substring of the job. Any other value means that
- the command must be a prefix of a stopped job.
- histchars
- Characters controlling history expansion and quick
- substitution. The first character is the history
- substitution character, usually `!'. The second is
- the `quick substitution' character, usually `^'. The
- third is the `history comment' character, usually `#'.
- HISTIGNORE
- A colon-separated list of patterns used to decide which
- commands should be saved on the history list.
- Execute conditional command.
- Returns a status of 0 or 1 depending on the evaluation of the conditional
- expression EXPRESSION. Expressions are composed of the same primaries used
- by the `test' builtin, and may be combined using the following operators:
- (EXPRESSION)
- Returns the value of EXPRESSION
- ! EXPRESSION

- True if EXPRESSION is false; else false
- EXPR1 && EXPR2
- True if both EXPR1 and EXPR2 are true; else false
- EXPR1 || EXPR2
- True if either EXPR1 or EXPR2 is true; else false
- When the `==' and` !=' operators are used, the string to the right of
- the operator is used as a pattern and pattern matching is performed.
- When the `=~' operator is used, the string to the right of the operator
- is matched as a regular expression.
- The && and || operators do not evaluate EXPR2 if EXPR1 is sufficient to
- determine the expression's value.
- Exit Status:
- 0 or 1 depending on value of EXPRESSION.
- Evaluate arithmetic expression.
- The EXPRESSION is evaluated according to the rules for arithmetic
- evaluation. Equivalent to "let EXPRESSION".
- Exit Status:
- Returns 1 if EXPRESSION evaluates to 0; returns 0 otherwise.
- Resume job in foreground.
- Equivalent to the JOB_SPEC argument to the `fg' command. Resume a
- stopped or background job. JOB_SPEC can specify either a job name
- or a job number. Following JOB_SPEC with a `&' places the job in
- the background, as if the job specification had been supplied as an
- argument to `bg'.
- Exit Status:
- Returns the status of the resumed job.
- Group commands as a unit.
- Run a set of commands in a group. This is one way to redirect an
- entire set of commands.
- Exit Status:
- Returns the status of the last command executed.
- Define shell function.
- Create a shell function named NAME. When invoked as a simple command,
- NAME runs COMMANDs in the calling shell's context. When NAME is invoked,
- the arguments are passed to the function as $1...$n, and the function's
- name is in $FUNCNAME.
- Exit Status:
- Returns success unless NAME is readonly.
- Create a coprocess named NAME.
- Execute COMMAND asynchronously, with the standard output and standard
- input of the command connected via a pipe to file descriptors assigned
- to indices 0 and 1 of an array variable NAME in the executing shell.
- The default NAME is "COPROC".
- Exit Status:
- Returns the exit status of COMMAND.
- Execute commands as long as a test does not succeed.
- Expand and execute COMMANDS as long as the final command in the
- `until' COMMANDS has an exit status which is not zero.
- Exit Status:
- Returns the status of the last command executed.
- Execute commands as long as a test succeeds.
- Expand and execute COMMANDS as long as the final command in the
- `while' COMMANDS has an exit status of zero.
- Exit Status:
- Returns the status of the last command executed.
- Execute commands based on conditional.
- The `if COMMANDS' list is executed. If its exit status is zero, then the
- `then COMMANDS' list is executed. Otherwise, each` elif COMMANDS' list is
- executed in turn, and if its exit status is zero, the corresponding
- `then COMMANDS' list is executed and the if command completes. Otherwise,
- the `else COMMANDS' list is executed, if present. The exit status of the
- entire construct is the exit status of the last command executed, or zero
- if no condition tested true.
- Exit Status:
- Returns the status of the last command executed.
- Execute commands based on pattern matching.
- Selectively execute COMMANDS based upon WORD matching PATTERN. The
- `|' is used to separate multiple patterns.
- Exit Status:
- Returns the status of the last command executed.
- Report time consumed by pipeline's execution.
- Execute PIPELINE and print a summary of the real time, user CPU time,
- and system CPU time spent executing PIPELINE when it terminates.

- Options:
- -p
- print the timing summary in the portable Posix format
- The value of the TIMEFORMAT variable is used as the output format.
- Exit Status:
- The return status is the return status of PIPELINE.
- Select words from a list and execute commands.
- The WORDS are expanded, generating a list of words. The
- set of expanded words is printed on the standard error, each
- preceded by a number. If `in WORDS' is not present,` in "$@"'
- is assumed. The PS3 prompt is then displayed and a line read
- from the standard input. If the line consists of the number
- corresponding to one of the displayed words, then NAME is set
- to that word. If the line is empty, WORDS and the prompt are
- redisplayed. If EOF is read, the command completes. Any other
- value read causes NAME to be set to null. The line read is saved
- in the variable REPLY. COMMANDS are executed after each selection
- until a break command is executed.
- Exit Status:
- Returns the status of the last command executed.
- Arithmetic for loop.
- Equivalent to
- ((EXP1))
- while ((EXP2)); do
- COMMANDS
- ((EXP3))
- EXP1, EXP2, and EXP3 are arithmetic expressions. If any expression is
- omitted, it behaves as if it evaluates to 1.
- Exit Status:
- Returns the status of the last command executed.
- Execute commands for each member in a list.
- The `for' loop executes a sequence of commands for each member in a
- list of items. If `in WORDS ...;' is not present, then` in "$@"' is
- assumed. For each element in WORDS, NAME is set to that element, and
- the COMMANDS are executed.
- Exit Status:
- Returns the status of the last command executed.
- Wait for job completion and return exit status.
- Waits for the process identified by ID, which may be a process ID or a
- job specification, and reports its termination status. If ID is not
- given, waits for all currently active child processes, and the return
- status is zero. If ID is a a job specification, waits for all processes
- in the job's pipeline.
- Exit Status:
- Returns the status of ID; fails if ID is invalid or an invalid option is
- given.
- Display or set file mode mask.
- Sets the user file-creation mask to MODE. If MODE is omitted, prints
- the current value of the mask.
- If MODE begins with a digit, it is interpreted as an octal number;
- otherwise it is a symbolic mode string like that accepted by chmod(1).
- Options:
- -p
- if MODE is omitted, output in a form that may be reused as input
- -S
- makes the output symbolic; otherwise an octal number is output
- Exit Status:
- Returns success unless MODE is invalid or an invalid option is given.
- Modify shell resource limits.
- Provides control over the resources available to the shell and processes
- it creates, on systems that allow such control.
- Options:
- -S
- use the `soft' resource limit
- -H
- use the `hard' resource limit
- -a
- all current limits are reported
- -b
- the socket buffer size
- -c
- the maximum size of core files created
- -d

- the maximum size of a process's data segment
- -e
- the maximum scheduling priority (`nice')
- -f
- the maximum size of files written by the shell and its children
- -i
- the maximum number of pending signals
- -l
- the maximum size a process may lock into memory
- -m
- the maximum resident set size
- -n
- the maximum number of open file descriptors
- -p
- the pipe buffer size
- -q
- the maximum number of bytes in POSIX message queues
- -r
- the maximum real-time scheduling priority
- -s
- the maximum stack size
- -t
- the maximum amount of cpu time in seconds
- -u
- the maximum number of user processes
- -v
- the size of virtual memory
- -x
- the maximum number of file locks
- If LIMIT is given, it is the new value of the specified resource; the
- special LIMIT values `soft',` hard', and `unlimited' stand for the
- current soft limit, the current hard limit, and no limit, respectively.
- Otherwise, the current value of the specified resource is printed. If
- no option is given, then -f is assumed.
- Values are in 1024-byte increments, except for -t, which is in seconds,
- -p, which is in increments of 512 bytes, and -u, which is an unscaled
- number of processes.
- Exit Status:
- Returns success unless an invalid option is supplied or an error occurs.
- Display information about command type.
- For each NAME, indicate how it would be interpreted if used as a
- command name.
- Options:
- -a
- display all locations containing an executable named NAME;
- includes aliases, builtins, and functions, if and only if
- the `-p' option is not also used
- -f
- suppress shell function lookup
- -P
- force a PATH search for each NAME, even if it is an alias,
- builtin, or function, and returns the name of the disk file
- that would be executed
- -p
- returns either the name of the disk file that would be executed,
- or nothing if `type -t NAME' would not return file'.
- -t
- output a single word which is one of `alias',` keyword',
- `function',` builtin', `file' or`', if NAME is an alias, shell
- reserved word, shell function, shell builtin, disk file, or not
- found, respectively
- Arguments:
- NAME
- Command name to be interpreted.
- Exit Status:
- Returns success if all of the NAMEs are found; fails if any are not found.
- Trap signals and other events.
- Defines and activates handlers to be run when the shell receives signals
- or other conditions.
- ARG is a command to be read and executed when the shell receives the
- signal(s) SIGNAL*SPEC. If ARG is absent (and a single SIGNAL*SPEC
- is supplied) or `-', each specified signal is reset to its original
- value. If ARG is the null string each SIGNAL_SPEC is ignored by the

- shell and by the commands it invokes.
- If a SIGNAL_SPEC is EXIT (0) ARG is executed on exit from the shell. If
- a SIGNAL_SPEC is DEBUG, ARG is executed before every simple command. If
- a SIGNAL_SPEC is RETURN, ARG is executed each time a shell function or a
- script run by the . or source builtins finishes executing. A SIGNAL_SPEC
- of ERR means to execute ARG each time a command's failure would cause the
- shell to exit when the -e option is enabled.
- If no arguments are supplied, trap prints the list of commands associated
- with each signal.
- Options:
- -l
- print a list of signal names and their corresponding numbers
- -p
- display the trap commands associated with each SIGNAL_SPEC
- Each SIGNAL_SPEC is either a signal name in or a signal number.
- Signal names are case insensitive and the SIG prefix is optional. A
- signal may be sent to the shell with "kill -signal $$".
- Exit Status:
- Returns success unless a SIGSPEC is invalid or an invalid option is given.
- Display process times.
- Prints the accumulated user and system times for the shell and all of its
- child processes.
- Exit Status:
- Always succeeds.
- Evaluate conditional expression.
- This is a synonym for the "test" builtin, but the last argument must
- be a literal `]', to match the opening`[.
- Evaluate conditional expression.
- Exits with a status of 0 (true) or 1 (false) depending on
- the evaluation of EXPR. Expressions may be unary or binary. Unary
- expressions are often used to examine the status of a file. There
- are string operators and numeric comparison operators as well.
- The behavior of test depends on the number of arguments. Read the
- bash manual page for the complete specification.
- File operators:
- -a FILE True if file exists.
- -b FILE True if file is block special.
- -c FILE True if file is character special.
- -d FILE True if file is a directory.
- -e FILE True if file exists.
- -f FILE True if file exists and is a regular file.
- -g FILE True if file is set-group-id.
- -h FILE True if file is a symbolic link.
- -L FILE True if file is a symbolic link.
- -k FILE True if file has its `sticky' bit set.
- -p FILE True if file is a named pipe.
- -r FILE True if file is readable by you.
- -s FILE True if file exists and is not empty.
- -S FILE True if file is a socket.
- -t FD True if FD is opened on a terminal.
- -u FILE True if the file is set-user-id.
- -w FILE True if the file is writable by you.
- -x FILE True if the file is executable by you.
- -O FILE True if the file is effectively owned by you.
- -G FILE True if the file is effectively owned by your group.
- -N FILE True if the file has been modified since it was last read.
- FILE1 -nt FILE2 True if file1 is newer than file2 (according to
- modification date).
- FILE1 -ot FILE2 True if file1 is older than file2.
- FILE1 -ef FILE2 True if file1 is a hard link to file2.
- String operators:
- -z STRING True if string is empty.
- -n STRING
- STRING True if string is not empty.
- STRING1 = STRING2
- True if the strings are equal.
- STRING1 != STRING2
- True if the strings are not equal.
- STRING1 < STRING2
- True if STRING1 sorts before STRING2 lexicographically.
- STRING1 > STRING2
- True if STRING1 sorts after STRING2 lexicographically.
- Other operators:

- -o OPTION True if the shell option OPTION is enabled.
- -v VAR
- True if the shell variable VAR is set
- ! EXPR True if expr is false.
- EXPR1 -a EXPR2 True if both expr1 AND expr2 are true.
- EXPR1 -o EXPR2 True if either expr1 OR expr2 is true.
- arg1 OP arg2 Arithmetic tests. OP is one of -eq, -ne,
- -lt, -le, -gt, or -ge.
- Arithmetic binary operators return true if ARG1 is equal, not-equal,
- less-than, less-than-or-equal, greater-than, or greater-than-or-equal
- than ARG2.
- Exit Status:
- Returns success if EXPR evaluates to true; fails if EXPR evaluates to
- false or an invalid argument is given.
- Suspend shell execution.
- Suspend the execution of this shell until it receives a SIGCONT signal.
- Unless forced, login shells cannot be suspended.
- Options:
- -f
- force the suspend, even if the shell is a login shell
- Exit Status:
- Returns success unless job control is not enabled or an error occurs.
- Execute commands from a file in the current shell.
- Read and execute commands from FILENAME in the current shell. The
- entries in $PATH are used to find the directory containing FILENAME.
- If any ARGUMENTS are supplied, they become the positional parameters
- when FILENAME is executed.
- Exit Status:
- Returns the status of the last command executed in FILENAME; fails if
- FILENAME cannot be read.
- Shift positional parameters.
- Rename the positional parameters $N+1,$N+2 ... to $1,$2 ... If N is
- not given, it is assumed to be 1.
- Exit Status:
- Returns success unless N is negative or greater than $#.
- Mark shell variables as unchangeable.
- Mark each NAME as read-only; the values of these NAMEs may not be
- changed by subsequent assignment. If VALUE is supplied, assign VALUE
- before marking as read-only.
- Options:
- -a
- refer to indexed array variables
- -A
- refer to associative array variables
- -f
- refer to shell functions
- -p
- display a list of all readonly variables and functions
- An argument of `--' disables further option processing.
- Exit Status:
- Returns success unless an invalid option is given or NAME is invalid.
- Set export attribute for shell variables.
- Marks each NAME for automatic export to the environment of subsequently
- executed commands. If VALUE is supplied, assign VALUE before exporting.
- Options:
- -f
- refer to shell functions
- -n
- remove the export property from each NAME
- -p
- display a list of all exported variables and functions
- An argument of `--' disables further option processing.
- Exit Status:
- Returns success unless an invalid option is given or NAME is invalid.
- Unset values and attributes of shell variables and functions.
- For each NAME, remove the corresponding variable or function.
- Options:
- -f
- treat each NAME as a shell function
- -v
- treat each NAME as a shell variable
- Without options, unset first tries to unset a variable, and if that fails,
- tries to unset a function.

- Some variables cannot be unset; also see `readonly'.
- Exit Status:
- Returns success unless an invalid option is given or a NAME is read-only.
- Set or unset values of shell options and positional parameters.
- Change the value of shell attributes and positional parameters, or
- display the names and values of shell variables.
- Options:
- -a Mark variables which are modified or created for export.
- -b Notify of job termination immediately.
- -e Exit immediately if a command exits with a non-zero status.
- -f Disable file name generation (globbing).
- -h Remember the location of commands as they are looked up.
- -k All assignment arguments are placed in the environment for a
- command, not just those that precede the command name.
- -m Job control is enabled.
- -n Read commands but do not execute them.
- -o option-name
- Set the variable corresponding to option-name:
- allexport same as -a
- braceexpand same as -B
- emacs use an emacs-style line editing interface
- errexit same as -e
- errtrace same as -E
- functrace same as -T
- hashall same as -h
- histexpand same as -H
- history enable command history
- ignoreeof the shell will not exit upon reading EOF
- interactive-comments
- allow comments to appear in interactive commands
- keyword same as -k
- monitor same as -m
- noclobber same as -C
- noexec same as -n
- noglob same as -f
- nolog currently accepted but ignored
- notify same as -b
- nounset same as -u
- onecmd same as -t
- physical same as -P
- pipefail the return value of a pipeline is the status of
- the last command to exit with a non-zero status,
- or zero if no command exited with a non-zero status
- posix change the behavior of bash where the default
- operation differs from the Posix standard to
- match the standard
- privileged same as -p
- verbose same as -v
- vi use a vi-style line editing interface
- xtrace same as -x
- -p Turned on whenever the real and effective user ids do not match.
- Disables processing of the $ENV file and importing of shell
- functions. Turning this option off causes the effective uid and
- gid to be set to the real uid and gid.
- -t Exit after reading and executing one command.
- -u Treat unset variables as an error when substituting.
- -v Print shell input lines as they are read.
- -x Print commands and their arguments as they are executed.
- -B the shell will perform brace expansion
- -C If set, disallow existing regular files to be overwritten
- by redirection of output.
- -E If set, the ERR trap is inherited by shell functions.
- -H Enable ! style history substitution. This flag is on
- by default when the shell is interactive.
- -P If set, do not follow symbolic links when executing commands
- such as cd which change the current directory.
- -T If set, the DEBUG trap is inherited by shell functions.
- -- Assign any remaining arguments to the positional parameters.
- If there are no remaining arguments, the positional parameters
- are unset.
-
    - Assign any remaining arguments to the positional parameters.
- The -x and -v options are turned off.

- Using + rather than - causes these flags to be turned off. The
- flags can also be used upon invocation of the shell. The current
- set of flags may be found in $-. The remaining n ARGs are positional
- parameters and are assigned, in order, to $1, $2, .. $n. If no
- ARGs are given, all shell variables are printed.
- Exit Status:
- Returns success unless an invalid option is given.
- Return from a shell function.
- Causes a function or sourced script to exit with the return value
- specified by N. If N is omitted, the return status is that of the
- last command executed within the function or script.
- Exit Status:
- Returns N, or failure if the shell is not executing a function or script.
- Read a line from the standard input and split it into fields.
- Reads a single line from the standard input, or from file descriptor FD
- if the -u option is supplied. The line is split into fields as with word
- splitting, and the first word is assigned to the first NAME, the second
- word to the second NAME, and so on, with any leftover words assigned to
- the last NAME. Only the characters found in $IFS are recognized as word
- delimiters.
- If no NAMEs are supplied, the line read is stored in the REPLY variable.
- Options:
- -a array
- assign the words read to sequential indices of the array
- variable ARRAY, starting at zero
- -d delim
- continue until the first character of DELIM is read, rather
- than newline
- -e
- use Readline to obtain the line in an interactive shell
- -i text
- Use TEXT as the initial text for Readline
- -n nchars
- return after reading NCHARS characters rather than waiting
- for a newline, but honor a delimiter if fewer than NCHARS
- characters are read before the delimiter
- -N nchars
- return only after reading exactly NCHARS characters, unless
- EOF is encountered or read times out, ignoring any delimiter
- -p prompt
- output the string PROMPT without a trailing newline before
- attempting to read
- -r
- do not allow backslashes to escape any characters
- -s
- do not echo input coming from a terminal
- -t timeout
- time out and return failure if a complete line of input is
- not read withint TIMEOUT seconds. The value of the TMOUT
- variable is the default timeout. TIMEOUT may be a
- fractional number. If TIMEOUT is 0, read returns success only
- if input is available on the specified file descriptor. The
- exit status is greater than 128 if the timeout is exceeded
- -u fd
- read from file descriptor FD instead of the standard input
- Exit Status:
- The return code is zero, unless end-of-file is encountered, read times out,
- or an invalid file descriptor is supplied as the argument to -u.
- Evaluate arithmetic expressions.
- Evaluate each ARG as an arithmetic expression. Evaluation is done in
- fixed-width integers with no check for overflow, though division by 0
- is trapped and flagged as an error. The following list of operators is
- grouped into levels of equal-precedence operators. The levels are listed
- in order of decreasing precedence.
- id++, id--
- variable post-increment, post-decrement
- ++id, --id
- variable pre-increment, pre-decrement
- unary minus, plus
- logical and bitwise negation
- exponentiation
- *, /, %*
- *multiplication, division, remainder*

- *addition, subtraction*
- <<, >>
- *left and right bitwise shifts*
- <=, >=, <, >
- *comparison*
- ==, !=
- *equality, inequality*
- *bitwise AND*
- *bitwise XOR*
- *bitwise OR*
- *logical AND*
- *logical OR*
- *expr ? expr : expr*
- *conditional operator*
- =, *=, /=, %=,
- +=, -=, <<=, >>=,
- &=, ^=, |=
- *assignment*
- *Shell variables are allowed as operands. The name of the variable*
- *is replaced by its value (coerced to a fixed-width integer) within*
- *an expression. The variable need not have its integer attribute*
- *turned on to be used in an expression.*
- *Operators are evaluated in order of precedence. Sub-expressions in*
- *parentheses are evaluated first and may override the precedence*
- *rules above.*
- *Exit Status:*
- *If the last ARG evaluates to 0, let returns 1; let returns 0 otherwise.*
- *Send a signal to a job.*
- *Send the processes identified by PID or JOBSPEC the signal named by*
- *SIGSPEC or SIGNUM. If neither SIGSPEC nor SIGNUM is present, then*
- *SIGTERM is assumed.*
- *Options:*
- *-s sig*
- *SIG is a signal name*
- *-n sig*
- *SIG is a signal number*
- *-l*
- *list the signal names; if arguments follow* `-l' they are</li> <li>assumed to be signal numbers for which names should be listed</li> <li>Kill is a shell builtin for two reasons: it allows job IDs to be used</li> <li>instead of process IDs, and allows processes to be killed if the limit</li> <li>on processes that you can create is reached.</li> <li>Exit Status:</li> <li>Returns success unless an invalid option is given or an error occurs.</li> <li>Remove jobs from current shell.</li> <li>Removes each JOBSPEC argument from the table of active jobs. Without</li> <li>any JOBSPECs, the shell uses its notion of the current job.</li> <li>Options:</li> <li>-a</li> <li>remove all jobs if JOBSPEC is not supplied</li> <li>-h</li> <li>mark each JOBSPEC so that SIGHUP is not sent to the job if the</li> <li>shell receives a SIGHUP</li> <li>-r</li> <li>remove only running jobs</li> <li>Exit Status:</li> <li>Returns success unless an invalid option or JOBSPEC is given.</li> <li>Display status of jobs.</li> <li>Lists the active jobs. JOBSPEC restricts output to that job.</li> <li>Without options, the status of all active jobs is displayed.</li> <li>Options:</li> <li>-l</li> <li>lists process IDs in addition to the normal information</li> <li>-n</li> <li>list only processes that have changed status since the last</li> <li>notification</li> <li>-p</li> <li>lists process IDs only</li> <li>-r</li> <li>restrict output to running jobs</li> <li>-s</li> <li>restrict output to stopped jobs</li> <li>If -x is supplied, COMMAND is run after all job specifications that</li> <li>appear in ARGS have been replaced with the process ID of that job's</li> <li>process group leader.</li> <li>Exit Status:</li> <li>Returns success unless an invalid option is given or an error occurs.</li> <li>If -x is used, returns the exit status of COMMAND.</li> <li>Display or manipulate the history list.</li> <li>Display the history list with line numbers, prefixing each modified</li> <li>entry with a`'. An argument of N lists only the last N entries.
- Options:
- -c
- clear the history list by deleting all of the entries
- -d offset
- delete the history entry at offset OFFSET.
- -a
- append history lines from this session to the history file
- -n
- read all history lines not already read from the history file
- -r
- read the history file and append the contents to the history
- -w
- write the current history to the history file
- and append them to the history list
- -p
- perform history expansion on each ARG and display the result
- without storing it in the history list
- -s
- append the ARGs to the history list as a single entry
- If FILENAME is given, it is used as the history file. Otherwise,
- if $HISTFILE has a value, that is used, else ~/.bash*history.*
- *If the $HISTTIMEFORMAT variable is set and not null, its value is used*

- *as a format string for strftime(3) to print the time stamp associated*
- *with each displayed history entry. No time stamps are printed otherwise.*
- *Exit Status:*
- *Returns success unless an invalid option is given or an error occurs.*
- *Display information about builtin commands.*
- *Displays brief summaries of builtin commands. If PATTERN is*
- *specified, gives detailed help on all commands matching PATTERN,*
- *otherwise the list of help topics is printed.*
- *Options:*
- *-d*
- *output short description for each topic*
- *-m*
- *display usage in pseudo-manpage format*
- *-s*
- *output only a short usage synopsis for each topic matching*
- *PATTERN*
- *Arguments:*
- *PATTERN*
- *Pattern specifiying a help topic*
- *Exit Status:*
- *Returns success unless PATTERN is not found or an invalid option is given.*
- *Remember or display program locations.*
- *Determine and remember the full pathname of each command NAME. If*
- *no arguments are given, information about remembered commands is displayed.*
- *Options:*
- *-d*
- *forget the remembered location of each NAME*
- *-l*
- *display in a format that may be reused as input*
- *-p pathname*
- *use PATHNAME is the full pathname of NAME*
- *-r*
- *forget all remembered locations*
- *-t*
- *print the remembered location of each NAME, preceding*
- *each location with the corresponding NAME if multiple*
- *NAMEs are given*
- *Arguments:*
- *NAME*
- *Each NAME is searched for in $PATH and added to the list*
- *of remembered commands.*
- *Exit Status:*
- *Returns success unless NAME is not found or an invalid option is given.*
- *Move jobs to the background.*
- *Place the jobs identified by each JOB*SPEC in the background, as if they
- had been started with `&'. If JOB_SPEC is not present, the shell's notion</li> <li>of the current job is used.</li> <li>Exit Status:</li> <li>Returns success unless job control is not enabled or an error occurs.</li> <li>Move job to the foreground.</li> <li>Place the job identified by JOB_SPEC in the foreground, making it the</li> <li>current job. If JOB_SPEC is not present, the shell's notion of the</li> <li>current job is used.</li> <li>Exit Status:</li> <li>Status of command placed in foreground, or failure if an error occurs.</li> <li>Display or execute commands from the history list. </li> <li>fc is used to list or edit and re-execute commands from the history list.</li> <li>FIRST and LAST can be numbers specifying the range, or FIRST can be a</li> <li>string, which means the most recent command beginning with that</li> <li>string.</li> <li>Options:</li> <li>-e ENAME</li> <li>select which editor to use. Default is FCEDIT, then EDITOR,</li> <li>then vi</li> <li>-l </li> <li>list lines instead of editing</li> <li>-n</li> <li>omit line numbers when listing</li> <li>-r</li> <li>reverse the order of the lines (newest listed first)</li> <li>With the` fc -s [pat=rep ...] [command]' format, COMMAND is
- re-executed after the substitution OLD=NEW is performed.
- A useful alias to use with this is r='fc -s', so that typing `r cc'</li> <li>runs the last command beginning with <code>cc' and typing</code>r' re-executes</li> <li>the last command.</li> <li>Exit Status:</li> <li>Returns success or status of executed command; non-zero if an error occurs.</li> <li>Exit a login shell.</li> <li>Exits a login shell with exit status N. Returns an error if not executed</li> <li>in a login shell.</li> <li>Exit the shell.</li> <li>Exits the shell with a status of N. If N is omitted, the exit status</li> <li>is that of the last command executed.</li> <li>Replace the shell with the given command.</li> <li>Execute COMMAND, replacing this shell with the specified program.</li> <li>ARGUMENTS become the arguments to COMMAND. If COMMAND is not specified,</li> <li>any redirections take effect in the current shell.</li> <li>Options:</li> <li>-a name</li> <li>pass NAME as the zeroth argument to COMMAND</li> <li>-c</li> <li>execute COMMAND with an empty environment</li> <li>-l</li> <li>place a dash in the zeroth argument to COMMAND</li> <li>If the command cannot be executed, a non-interactive shell exits, unless</li> <li>the shell option` execfail' is set.
- Exit Status:
- Returns success unless COMMAND is not found or a redirection error occurs.
- Parse option arguments.
- Getopts is used by shell procedures to parse positional parameters
- as options.
- OPTSTRING contains the option letters to be recognized; if a letter
- is followed by a colon, the option is expected to have an argument,
- which should be separated from it by white space.
- Each time it is invoked, getopts will place the next option in the
- shell variable $name, initializing name if it does not exist, and

- the index of the next argument to be processed into the shell
- variable OPTIND. OPTIND is initialized to 1 each time the shell or
- a shell script is invoked. When an option requires an argument,
- getopts places that argument into the shell variable OPTARG.
- getopts reports errors in one of two ways. If the first character
- of OPTSTRING is a colon, getopts uses silent error reporting. In
- this mode, no error messages are printed. If an invalid option is
- seen, getopts places the option character found into OPTARG. If a
- required argument is not found, getopts places a ':' into NAME and
- sets OPTARG to the option character found. If getopts is not in
- silent mode, and an invalid option is seen, getopts places '?' into
- NAME and unsets OPTARG. If a required argument is not found, a '?'
- is placed in NAME, OPTARG is unset, and a diagnostic message is
- printed.
- If the shell variable OPTERR has the value 0, getopts disables the
- printing of error messages, even if the first character of
- OPTSTRING is not a colon. OPTERR has the value 1 by default.
- Getopts normally parses the positional parameters ($0 - $9), but if
- more arguments are given, they are parsed instead.
- Exit Status:
- Returns success if an option is found; fails if the end of options is
- encountered or an error occurs.
- Execute arguments as a shell command.
- Combine ARGs into a single string, use the result as input to the shell,
- and execute the resulting commands.
- Exit Status:
- Returns exit status of command or success if command is null.
- Enable and disable shell builtins.
- Enables and disables builtin shell commands. Disabling allows you to
- execute a disk command which has the same name as a shell builtin
- without using a full pathname.
- Options:
- -a
- print a list of builtins showing whether or not each is enabled
- -n
- disable each NAME or display a list of disabled builtins
- -p
- print the list of builtins in a reusable format
- -s
- print only the names of Posix `special' builtins</li> <li>Options controlling dynamic loading:</li> <li>-f</li> <li>Load builtin NAME from shared object FILENAME</li> <li>-d</li> <li>Remove a builtin loaded with -f</li> <li>Without options, each NAME is enabled.</li> <li>To use the` test' found in $PATH instead of the shell builtin
- version, type `enable -n test'.</li> <li>Exit Status:</li> <li>Returns success unless NAME is not a shell builtin or an error occurs.</li> <li>Write arguments to the standard output.</li> <li>Display the ARGs on the standard output followed by a newline.</li> <li>Options:</li> <li>-n</li> <li>do not append a newline</li> <li>-e</li> <li>enable interpretation of the following backslash escapes</li> <li>-E</li> <li>explicitly suppress interpretation of backslash escapes</li> <li>` echo' interprets the following backslash-escaped characters:
- \a
- alert (bell)
- \b
- backspace
- \c
- suppress further output
- \e
- escape character
- \f
- form feed
- \n
- new line
- \r
- carriage return
- \t
- horizontal tab
- \v
- vertical tab
- \
- backslash
- \0nnn
- the character whose ASCII code is NNN (octal). NNN can be
- 0 to 3 octal digits
- \xHH
- the eight-bit character whose value is HH (hexadecimal). HH
- can be one or two hex digits
- Exit Status:

- Returns success unless a write error occurs.
- Define local variables.
- Create a local variable called NAME, and give it VALUE. OPTION can
- be any option accepted by `declare'.</li> <li>Local variables can only be used within a function; they are visible</li> <li>only to the function where they are defined and its children.</li> <li>Exit Status:</li> <li>Returns success unless an invalid option is supplied, an error occurs,</li> <li>or the shell is not executing a function.</li> <li>Set variable values and attributes.</li> <li>Obsolete. See` help declare'.
- Set variable values and attributes.
- Declare variables and give them attributes. If no NAMEs are given,
- display the attributes and values of all variables.
- Options:
- -f
- restrict action or display to function names and definitions
- -F
- restrict display to function names only (plus line number and
- source file when debugging)
- -g
- create global variables when used in a shell function; otherwise
- ignored
- -p
- display the attributes and value of each NAME
- Options which set attributes:
- -a
- to make NAMEs indexed arrays (if supported)
- -A
- to make NAMEs associative arrays (if supported)
- -i
- to make NAMEs have the `integer' attribute</li> <li>-l</li> <li>to convert NAMEs to lower case on assignment</li> <li>-r</li> <li>to make NAMEs readonly</li> <li>-t</li> <li>to make NAMEs have the` trace' attribute
- -u
- to convert NAMEs to upper case on assignment
- -x
- to make NAMEs export
- Using `+' instead of` -' turns off the given attribute.
- Variables with the integer attribute have arithmetic evaluation (see
- the `let' command) performed when the variable is assigned a value.</li> <li>When used in a function, <code>declare' makes NAMEs local, as with the</code>local'</li> <li>command. The` -g' option suppresses this behavior.
- Exit Status:
- Returns success unless an invalid option is supplied or an error occurs.
- Execute a simple command or display information about commands.
- Runs COMMAND with ARGS suppressing shell function lookup, or display
- information about the specified COMMANDs. Can be used to invoke commands
- on disk when a function with the same name exists.
- Options:
- -p
- use a default value for PATH that is guaranteed to find all of
- the standard utilities
- -v
- print a description of COMMAND similar to the `type' builtin</li> <li>-V</li> <li>print a more verbose description of each COMMAND</li> <li>Exit Status:</li> <li>Returns exit status of COMMAND, or failure if COMMAND is not found.</li> <li>Return an unsuccessful result.</li> <li>Exit Status:</li> <li>Always fails.</li> <li>Return a successful result.</li> <li>Exit Status:</li> <li>Always succeeds.</li> <li>Null command.</li> <li>No effect; the command does nothing.</li> <li>Exit Status:</li> <li>Always succeeds.</li> <li>Print the name of the current working directory.</li> <li>Options:</li> <li>-L</li> <li>print the value of $PWD if it names the current working</li> <li>directory</li> <li>-P</li> <li>print the physical directory, without any symbolic links</li> <li>By default, <code>pwd' behaves as if</code>-L' were specified.</li> <li>Exit Status:</li> <li>Returns 0 unless an invalid option is given or the current directory</li> <li>cannot be read.</li> <li>Change the shell working directory.</li> <li>Change the current directory to DIR. The default DIR is the value of the</li> <li>HOME shell variable.</li> <li>The variable CDPATH defines the search path for the directory containing</li> <li>DIR. Alternative directory names in CDPATH are separated by a colon (:).</li> <li>A null directory name is the same as the current directory. If DIR begins</li> <li>with a slash (/), then CDPATH is not used.</li> <li>If the directory is not found, and the shell option` cdable*vars' is set,*
- *the word is assumed to be a variable name. If that variable has a value,*
- *its value is used for DIR.*
- *Options:*
- *-L*
- *force symbolic links to be followed*
- *-P*
- *use the physical directory structure without following symbolic*
- *-e*
- *if the -P option is supplied, and the current working directory*
- *cannot be determined successfully, exit with a non-zero status*
- *The default is to followsymbolic links, as if `-L' were specified.</li> <li>Exit Status:</li> <li>Returns 0 if the directory is changed, and if $PWD is set successfully when</li> <li>-P is used; non-zero otherwise.</li> <li>Return the context of the current subroutine call.</li> <li>Without EXPR, returns "$line $filename". With EXPR, returns</li> <li>"$line $subroutine $filename"; this extra information can be used to</li> <li>provide a stack trace.</li> <li>The value of EXPR indicates how many call frames to go back before the</li> <li>current one; the top frame is frame 0.</li> <li>Exit Status:</li> <li>Returns 0 unless the shell is not executing a shell function or EXPR</li> <li>is invalid.</li> <li>Execute shell builtins.</li> <li>Execute*

SHELL-BUILTIN with arguments ARGs without performing command</li> <li>lookup. This is useful when you wish to reimplement a shell builtin</li> <li>as a shell function, but need to execute the builtin within the function.</li> <li>Exit Status:</li> <li>Returns the exit status of SHELL-BUILTIN, or false if SHELL-BUILTIN is</li> <li>not a shell builtin..</li> <li>Resume for, while, or until loops.</li> <li>Resumes the next iteration of the enclosing FOR, WHILE or UNTIL loop.</li> <li>If N is specified, resumes the Nth enclosing loop.</li> <li>Exit Status:</li> <li>The exit status is 0 unless N is not greater than or equal to 1.</li> <li>Exit for, while, or until loops.</li> <li>Exit a FOR, WHILE or UNTIL loop. If N is specified, break N enclosing</li> <li>loops.</li> <li>Exit Status:</li> <li>The exit status is 0 unless N is not greater than or equal to 1.</li> <li>Set Readline key bindings and variables.</li> <li>Bind a key sequence to a Readline function or a macro, or set a</li> <li>Readline variable. The non-option argument syntax is equivalent to</li> <li>that found in ~/.inputrc, but must be passed as a single argument:</li> <li>e.g., bind '"\C-x\C-r": re-read-init-file'.</li> <li>Options:</li> <li>-m keymap Use KEYMAP as the keymap for the duration of this</li> <li>command. Acceptable keymap names are emacs,</li> <li>emacs-standard, emacs-meta, emacs-ctlx, vi, vi-move,</li> <li>vi-command, and vi-insert.</li> <li>-l List names of functions.</li> <li>-P List function names and bindings.</li> <li>-p List functions and bindings in a form that can be</li> <li>reused as input.</li> <li>-S List key sequences that invoke macros and their values</li> <li>-s List key sequences that invoke macros and their values</li> <li>in a form that can be reused as input.</li> <li>-V List variable names and values</li> <li>-v List variable names and values in a form that can</li> <li>be reused as input.</li> <li>-q function-name Query about which keys invoke the named function.</li> <li>-u function-name Unbind all keys which are bound to the named function.</li> <li>-r keyseq Remove the binding for KEYSEQ.</li> <li>-f filename Read key bindings from FILENAME.</li> <li>-x keyseq:shell-command</li> <li>Cause SHELL-COMMAND to be executed when</li> <li>KEYSEQ is entered.</li> <li>Exit Status:</li> <li>bind returns 0 unless an unrecognized option is given or an error occurs.</li> <li>Remove each NAME from the list of defined aliases.</li> <li>Options:</li> <li>-a</li> <li>remove all alias definitions.</li> <li>Return success unless a NAME is not an existing alias.</li> <li>Define or display aliases.</li> <li>Without arguments, alias' prints the list of aliases in the reusable

- form alias NAME=VALUE' on standard output.</li> <li>Otherwise, an alias is defined for each NAME whose VALUE is given.</li> <li>A trailing space in VALUE causes the next word to be checked for</li> <li>alias substitution when the alias is expanded.</li> <li>Options:</li> <li>-p</li> <li>Print all defined aliases in a reusable format</li> <li>Exit Status:</li> <li>alias returns true unless a NAME is supplied for which no alias has been</li> <li>defined.</li> <li>alias [-p] [name[=value] ...]</li> <li>unalias</li> <li>unalias [-a] name [name ...]</li> <li>bind [-lpvsPVS] [-m keymap] [-f filename] [-q name] [-u name] [-r keyseq] [-x keyseq:shell-command] </li> <li>break [n]</li> <li>continue</li> <li>continue [n]</li> <li>builtin [shell-builtin [arg ...]]</li> <li>caller</li> <li>caller [expr]</li> <li>cd [-L|[-P [-e]]] [dir]</li> <li>pwd [-LP]</li> <li>command [-pVv] command [arg ...]</li> <li>declare</li> <li>declare [-aAfFgilrtux] [-p] [name[=value] ...]</li> <li>typeset</li> <li>typeset [-aAfFgilrtux] [-p] name[=value] ...</li> <li>local [option] name[=value] ...</li> <li>echo [-neE] [arg ...]</li> <li>enable</li> <li>enable [-a] [-dnps] [-f filename] [name ...]</li> <li>eval [arg ...]</li> <li>getopts</li> <li>getopts optstring name [arg]</li> <li>exec [-cl] [-a name] [command [arguments ...]] [redirection ...] </li> <li>exit [n]</li> <li>logout [n]</li> <li>fc [-e ename] [-lnr] [first] [last] or fc -s [pat=rep] [command]</li> <li>fg [job_spec]</li> <li>bg [job_spec ...]</li> <li>hash [-lr] [-p pathname] [-dt] [name ...]</li> <li>help [-dms] [pattern ...]</li> <li>history [-c] [-d offset] [n] or history -anrw [filename] or history -ps arg [arg...]</li> <li>jobs [-lnprs] [jobspec ...] or jobs -x command [args]</li> <li>disown</li> <li>disown [-h] [-ar] [jobspec ...]</li> <li>kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l [sigspec]</li> <li>let arg [arg ...]</li> <li>read [-ers] [-a array] [-d delim] [-i text] [-n nchars] [-N nchars] [-p prompt] [-t timeout] [-u fd]</li> <li>return</li> <li>return [n]</li> <li>set [-abefhkmnptuvxBCHP] [-o option-name] [--] [arg ...]</li> <li>unset [-f] [-v] [name ...]</li> <li>export</li> <li>export [-fn] [name[=value] ...] or export -p</li> <li>readonly</li> <li>readonly [-aAf] [name[=value] ...] or readonly -p</li> <li>shift [n]</li> <li>source</li> <li>source filename [arguments]</li> <li>. filename [arguments]</li> <li>suspend</li> <li>suspend [-f]</li> <li>test [expr]</li> <li>[arg...]</li> <li>trap [-lp] [[arg] signal_spec ...]</li> <li>type [-afptP] name [name ...]</li> <li>ulimit</li> <li>ulimit [-SHacdefilmnpqrstuvx] [limit]</li> <li>umask [-p] [-S] [mode] </li> <li>wait [id]</li> <li>for NAME [in WORDS ...] ; do COMMANDS; done</li> <li>for ((exp1; exp2; exp3)); do COMMANDS; done</li> <li>select NAME [in WORDS ... ;] do COMMANDS; done</li> <li>time [-p] pipeline</li> <li>case WORD in [PATTERN [| PATTERN]...) COMMANDS ;;]... esac</li> <li>if COMMANDS; then COMMANDS; [elif COMMANDS; then COMMANDS;]... [else COMMANDS;] fi</li> <li>while COMMANDS; do COMMANDS; done</li> <li>until COMMANDS; do COMMANDS; done</li> <li>coproc [NAME] command [redirections]</li> <li>function name { COMMANDS ; } or name () { COMMANDS ; }</li> <li>{ ... }</li> <li>job_spec [&]</li> <li>((...))</li> <li>((expression))</li> <li>[[...]]</li> <li>[[expression]]</li> <li>variables</li> <li>variables - Names and meanings of some shell variables</li> <li>pushd [-n] [+N | -N | dir]</li> <li>popd [-n] [+N | -N]</li> <li>dirs [-clpv] [+N] [-N]</li> <li>shopt [-pqsu] [-o] [optname ...]</li> <li>printf</li> <li>printf [-v var] format [arguments]</li> <li>complete [-abcdefgjksuv] [-pr] [-DE] [-o option] [-A action] [-G globpat] [-W wordlist] [-F functio</li> <li>compgen</li> <li>compgen [-abcdefgjksuv] [-o option] [-A action] [-G globpat] [-W wordlist] [-F function] [-C comma</li> <li>compopt</li> <li>compopt [-o|+o option] [-DE] [name ...]</li> <li>mapfile</li> <li>mapfile [-n count] [-O origin] [-s count] [-t] [-u fd] [-C callback] [-c quantum] [array]</li> <li>readarray</li> <li>readarray [-n count] [-O origin] [-s count] [-t] [-u fd] [-C callback] [-c quantum] [array]</li> <li>alias </li> <li>%s': invalid alias name
- line editing not enabled
- bindbuiltin
- lvpVPsSf:q:u:m:r:x:
- %s': invalid keymap name</li> <li>%s: cannot read: %s</li> <li>%s': unknown function name
- %s is not bound to any keys.
- %s can be invoked via
- "%s"%s
- %s': cannot unbind</li> <li>only meaningful in a <code>for',</code>while', orunti' loop
- loop count
- %s %s %s
- OLDPWD=
- HOME not set
- OLDPWD not set
- CDPATH
- /bin:/usr/bin:/sbin:/usr/sbin
- commandbuiltin
- %s: usage:
- %s: numeric argument required
- %s: not found
- %s: invalid option name
- invalid octal number
- invalid hex number
- %s: invalid signal specification
- %s': not a pid or valid job spec</li> <li>%s: %s out of range</li> <li>%s: no such job</li> <li>%s: no job control</li> <li>%s: restricted</li> <li>%s: not a shell builtin</li> <li>write error: %s</li> <li>error setting terminal attributes: %s</li> <li>error getting terminal attributes:

%s</li> <li>%s: error retrieving current directory: %s: %s</li> <li>%s: ambiguous job spec</li> <li>SIGJUNK</li> <li>Unknown</li> <li>%2d) %s</li> <li>+acfgilprtuxAF</li> <li>cannot use -f' to make functions

- %s %d %s
- %s: cannot destroy array variables in this way
- %s: cannot convert associative to indexed array
- can only be used in a function
- adnpsf:
- enable %s
- enable -n %s
- cannot open shared object %s: %s
- struct
- cannot find %s in shared object %s: %s
- %s: not dynamically loaded
- %s: cannot delete: %s
- %s: not a regular file
- %s: file is too large
- evalfile
- parseandexecute top
- pedispose
- parseandexecute
- parsestring top
- parsestring
- %s: cannot execute: %s
- ~/.bashlogout
- /etc/bash/bashlogout
- There are stopped jobs.
- There are running jobs.
- not login shell: use exit'</li> <li>logout</li> <li>${FCEDIT:-${EDITOR:-ed}}</li> <li>${FCEDIT:-${EDITOR:-vi}}</li> <li>:e:lnrs</li> <li>no command found</li> <li>history specification</li> <li>bash-fc</li> <li>%s: cannot open temp file: %s</li> <li>fc builtin</li> <li>current</li> <li>job %d started without job control</li> <li>builtin hash -p %s %s</li> <li>hashing disabled</li> <li>dlp:rt</li> <li>command</li> <li>%s: hash table empty</li> <li>%s: cannot open: %s</li> <li>These shell commands are defined internally. Type help' to see this list.
- Type help name' to find out more about the function name'.
- Use info bash' to find out more about the shell in general.</li> <li>Use <code>man -k' or</code>info' to find out more about commands not in this list.</li> <li>A star (*) next to a name means that the command is disabled.</li> <li>Shell commands matching keyword
- Shell commands matching keywords </li> <li>%*s%s - </li> <li>SYNOPSIS</li> <li>DESCRIPTION</li> <li>SEE ALSO</li> <li>%*sbash(1)</li> <li>IMPLEMENTATION</li> <li>no help topics match <code>%s'. Try</code>help help' or <code>man -k %s' or</code>info %s'.</li> <li>acd:npsrw</li> <li>cannot use more than one of -anrw</li> <li>%s: history expansion failed</li> <li>history position</li> <li>%5d%c %s%s</li> <li>no other options allowed with -x'
- lpnxrs
- jobsbuiltin
- Unknown error
- (%ld) - %s
- %s: arguments must be process or job IDs
- MAPFILE
- %s: invalid file descriptor specification
- %d: invalid file descriptor: %s
- %s: invalid line count
- %s: invalid array origin
- %s: invalid callback quantum
- u:n:O:tC:c:s:
- internal error: getting variable name
- empty array variable name
- %s: not an indexed array
- %s %d %s
- directory stack empty
- directory stack index
-
- %2d %s
- %2d %s
- no other directory
- dirstack
- %s: invalid timeout specification
- ersa:d:i:n:p:t:u:N:
- readbuiltin
- read error: %d: %s
- can only return' from a function or sourced script</li> <li>editing-mode</li> <li>set %co %s</li> <li>allexport</li> <li>SHELLOPTS</li> <li>cannot simultaneously unset a function and a variable</li> <li>%s: cannot unset</li> <li>%s: cannot unset: readonly %s</li> <li>%s: not an array variable</li> <li>braceexpand</li> <li>errexit</li> <li>errtrace</li> <li>functrace</li> <li>hashall</li> <li>histexpand</li> <li>interactive-comments</li> <li>keyword</li> <li>monitor</li> <li>noclobber</li> <li>noexec</li> <li>noglob</li> <li>notify</li> <li>nounset</li> <li>onecmd</li> <li>physical</li> <li>pipefail</li> <li>privileged</li> <li>xtrace</li> <li>declare -%s </li> <li>%s -%s </li> <li>%s: not a function</li> <li>shift count</li> <li>filename argument required</li> <li>%s: file not found</li> <li>cannot suspend</li> <li>cannot suspend a login shell</li> <li>trap -- %s %d</li> <li>trap -- %s %s</li> <li>%s is aliased to %s'
- alias %s=%s

- %s is a shell keyword
- %s is a function
- %s is a shell builtin
- %s is %s
- %s is hashed (%s)
- (%s, -%c)
- (-%c)
- %-20s %16s
- unlimited
- %s: cannot get limit: %s
- `%c': bad command</li> <li>%s: cannot modify limit: %s</li> <li>core file size</li> <li>blocks</li> <li>data seg size</li> <li>kbytes</li> <li>scheduling priority</li> <li>file size</li> <li>pending signals</li> <li>max locked memory</li> <li>max memory size</li> <li>open files</li> <li>pipe size</li> <li>512 bytes</li> <li>POSIX message queues</li> <li>real-time priority</li> <li>stack size</li> <li>cpu time</li> <li>seconds</li> <li>max user processes</li> <li>virtual memory</li> <li>file locks</li> <li>u=%s,g=%s,o=%s</li> <li> %c': invalid symbolic mode operator
- `%c': invalid symbolic mode character</li> <li>umask%s </li> <li>OPTARG</li> <li>autocd</li> <li>%s: invalid shell option name</li> <li>shopt %s %s</li> <li>BASHOPTS</li> <li>cannot set and unset shell options simultaneously</li> <li>cdable_vars</li> <li>cdspell</li> <li>checkhash</li> <li>checkjobs</li> <li>checkwinsize</li> <li>cmdhist</li> <li>compat31</li> <li>compat32</li> <li>compat40</li> <li>compat41</li> <li>direxpand</li> <li>dirspell</li> <li>dotglob</li> <li>execfail</li> <li>expand_aliases</li> <li>extdebug</li> <li>extglob</li> <li>extquote</li> <li>failglob</li> <li>force_fignore</li> <li>globstar</li> <li>gnu_errfmt</li> <li>histappend</li> <li>histreedit</li> <li>histverify</li> <li>hostcomplete</li> <li>huponexit</li> <li>interactive_comments</li> <li>lastpipe</li> <li>lithist</li> <li>mailwarn</li> <li>noemptycmd_completion</li> <li>nocaseglob</li> <li>nocasematch</li> <li>nullglob</li> <li>progcomp</li> <li>promptvars</li> <li>restricted_shell</li> <li>shift_verbose</li> <li>sourcepath</li> <li>xpg_echo</li> <li>missing hex digit for \x</li> <li>missing unicode digit for \%c</li> <li>warning: %s: %s</li> <li>#'-+ 0</li> <li>hjlLtz</li> <li> %s': missing format character
- `%c': invalid time format specification
- %s: illegal option -- %c
- %s: option requires an argument -- %c
- bashdefault
- abcdefgjko:prsuvA:G:W:P:S:X:F:C:DE
- %s: invalid action name
- complete
- -o %s
- dirnames
- filenames
- nospace
- plusdirs
- -A %s
- arrayvar
- binding
- enabled
- helptopic
- hostname
- running
- setopt
- signal
- %s %s
- %s: no completion specification
- compopt
- +o %s
- warning: -F option may not work as you expect
- warning: -C option may not work as you expect
- not currently executing completion function
- service
- xdigit
- backspace
- vertical-tab
- form-feed
- carriage-return
- exclamation-mark
- quotation-mark
- number-sign
- dollar-sign
- percent-sign
- ampersand
- apostrophe
- left-parenthesis
- right-parenthesis
- asterisk
- plus-sign
- hyphen
- hyphen-minus
- period
- full-stop
- semicolon

- less-than-sign
- equals-sign
- greater-than-sign
- question-mark
- commercial-at
- left-square-bracket
- backslash
- reverse-solidus
- right-square-bracket
- circumflex
- circumflex-accent
- underscore
- grave-accent
- left-brace
- left-curly-bracket
- vertical-line
- right-brace
- right-curly-bracket
- BASH*REMATCH*
- */dev/std*
- *%s: bad network path specification*
- *socket*
- *connect*
- *%ldm%d.%03ds*
- *sh*makepath
- sh*realpath*
- *TMPDIR*
- */var/tmp*
- */usr/tmp*
- *%s/%s-%lu*
- *%s/cur*
- *%s/tmp*
- *%s/new*
- *invalid base*
- *0123456789abcdef*
- *0123456789ABCDEF*
- *.shstrtab*
- *.interp*
- *.gnu.hash*
- *.dynsym*
- *.dynstr*
- *.gnu.version*
- *.gnu.versionr*
- .rela.dyn
- .rela.plt
- .rodata
- .eh*frame*hdr
- .eh*frame*
- *.init*array
- .fini_array
- .dynamic
- .note.ABI-tag

UTF-8 Strings

No detected.

UTF-16LE Strings

- fh#kPp
- <=%?>
- +,-./<
- #$%&'123+,-./:
- ;#$%&'
- 67@AEw
- !XYZ[</li>
- #$x%&'()*+,
- -./012

Extracted URLs

- http://gnu.org/licenses/gpl.html

Base64 Strings

- Possible Base64 Encoded String: d0d0d0d0d0d0d0d0d0d0d0d0d0d0d0d0 -> Decoded: wGtwGtwGtwGtwGtwGtwGtwGt

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File Entropy Analysis

- **Entropy**: `6.28`
- **Status**: This file is likely packed or encrypted.

VirusTotal Analysis

Scan Date: 2021-08-09 02:38:40

Scan Results

- Malicious: 0
- Suspicious: 0
- Undetected: 60
- Harmless: 0
- Timeout: 0
- Confirmed-timeout: 0
- Failure: 0
- Type-unsupported: 14

Community Votes

- Harmless: 0
- Malicious: 0

Detailed Engine Results

Engine Name	Detection Category	Detection
No results	No results	No results